Engineering Change Notice(ECN)

ECN-D-2607-02(MISC IMMERSION TRACK 개선 건)

FA R&D ME2

Written	Review	Approval
이준환	-	최동수
06/29	-	06/30

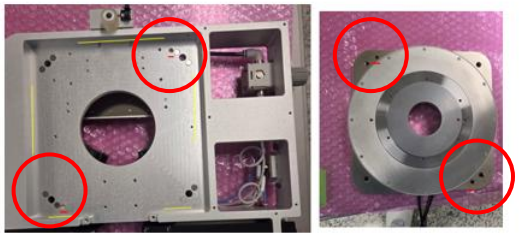
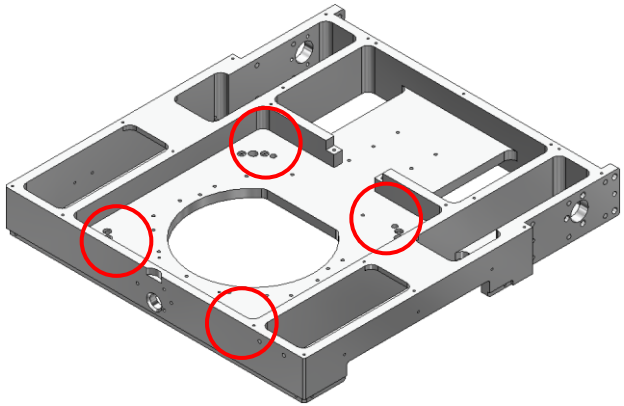
NINEBELL

1. Engineering Change Request

N/A

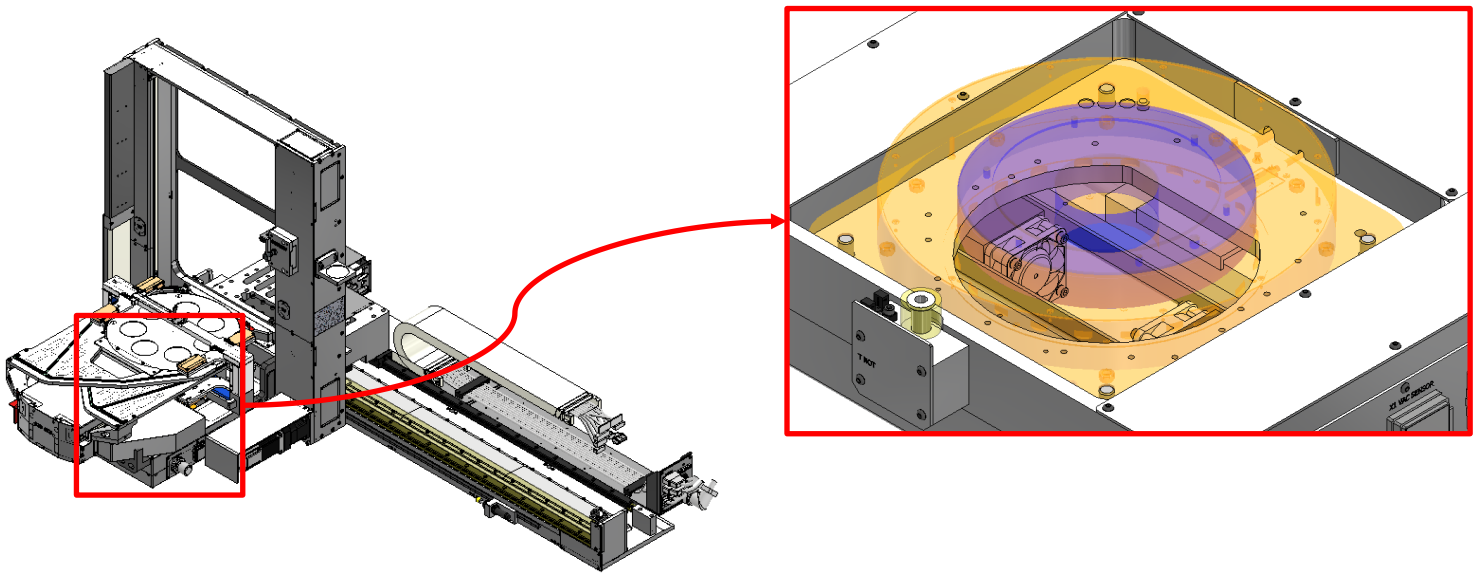
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #2 ~ #11	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	DD motor pin holes are machined for the as-drawn robots (#2, #4, #5, #6, #11). The pin hole locations do not match the mirrored robots (#2, #4, #5, #6, #11).		

변경 사항 Changes	변경 전(Before)	변경 후(After)
	1) ESR T BASE  1. 노랑색표시 4면 1mm필러추가하여 DD모터 고정하여 진행 협의 2. 다음 프로젝트 도면수정예정	
상세내용 Detail	2 DD motor pin holes are used diagonally. #4,#5,#6 P/N: TM01M-T100-M001-0 #7,#8,#9 P/N: TM01M-T100-M001-0_M #2,#11 P/N: TM01S-T100-M001-0 #3,#10 P/N: TM01S-T100-M001-0_M	Changed the number of DD motor pin holes from 2 to 4 for common use in both as-drawn and mirrored types. #4,#5,#6 P/N: TM01M-T100-M001-1 #7,#8,#9 P/N: TM01M-T100-M001-1_M #2,#11 P/N: TM01S-T100-M001-1 #3,#10 P/N: TM01S-T100-M001-1_M

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

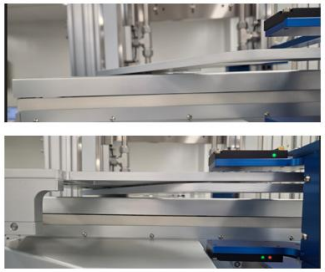
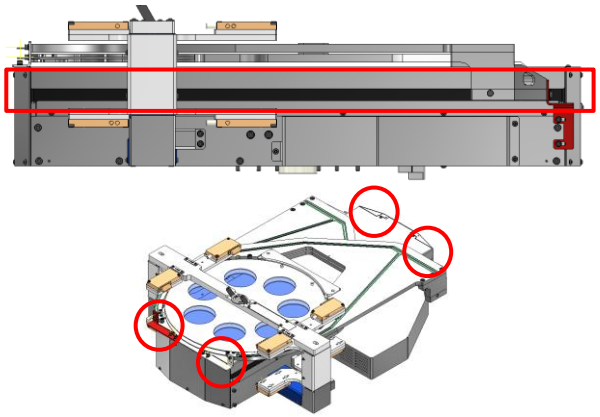
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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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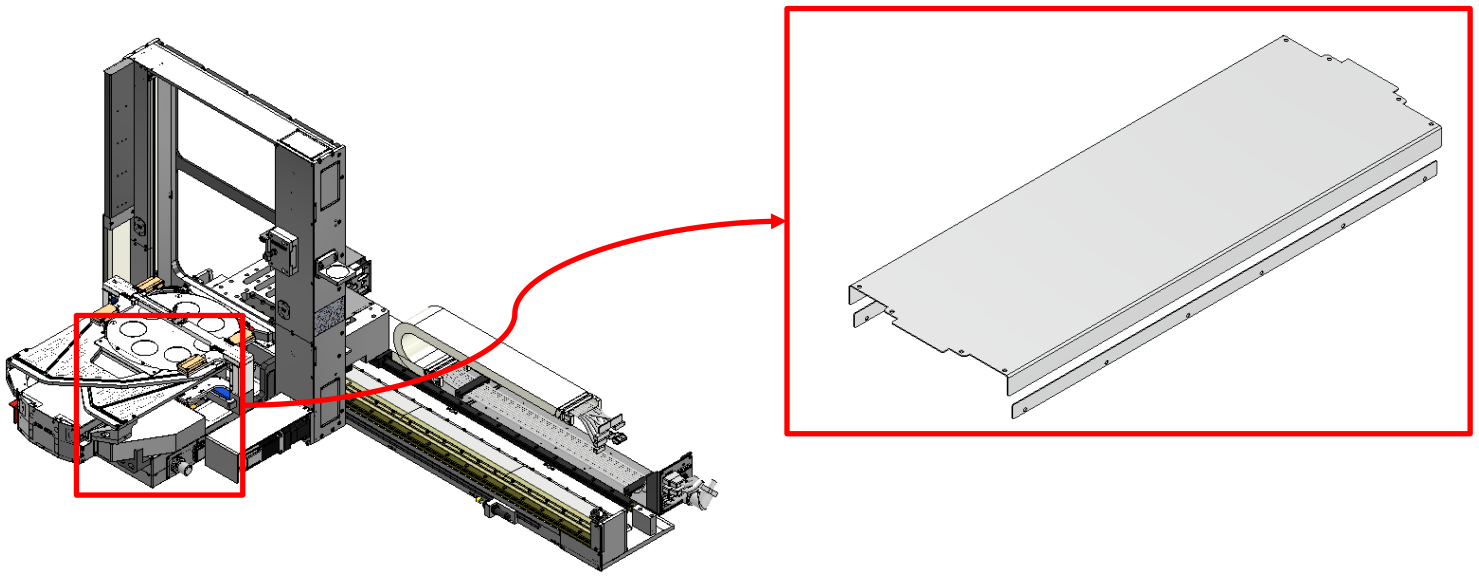
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-03-7813 IMMERSION TRACK PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1 ~ #12	적용 범위 Scope	■ 신규 발주분 (New order) ■ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO. ECN NO.	N/A ECN-D-2607-02
변경 상세 사유 Detailed reason for change	A gap was found between the X-axis sealing belt and the cover on some robots.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
	3) X AXIS SEALING BELT  • X AXIS 씰링벨트 커버 밴딩 및 벨트 텐션으로 X AXIS 구동 시 튐 발생 • 벨트텐션을 높여서 (45에서 47정도) 보완은 하였으나 현상은 남아있음	
상세내용 Detail	Covering gap for the X-axis sealing belt: 0.5mm Top cover fixing bolts: 4 ea #1 P/N: TM01E-X600-P009-0, TM01E-X600-P040-0 #4,#5,#6 P/N: TM01M-X100-P009-0, TM01M-X100-P001-0 #7,#8,#9 P/N: TM01M-X100-P009-0, TM01M-X100-P001-0 #2,#11 P/N: TM01M-X100-P009-0, TM01M-X100-P001-0 #3,#10 P/N: TM01M-X100-P009-0, TM01M-X100-P001-0 #12 P/N: TM01I-X600-P009-0, TM01I-X600-P040-0	Covering gap for the X-axis sealing belt changed: 0.5mm -> 1.0mm Top cover fixing bolts increased: 4 ea -> 8 ea #1 P/N: TM01E-X600-P009-1, TM01E-X600-P040-1 #4,#5,#6 P/N: TM01M-X100-P009-1, TM01M-X100-P001-1 #7,#8,#9 P/N: TM01M-X100-P009-1, TM01M-X100-P001-1 #2,#11 P/N: TM01M-X100-P009-1, TM01M-X100-P001-1 #3,#10 P/N: TM01M-X100-P009-1, TM01M-X100-P001-1 #12 P/N: TM01I-X600-P009-1, TM01I-X600-P040-1

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

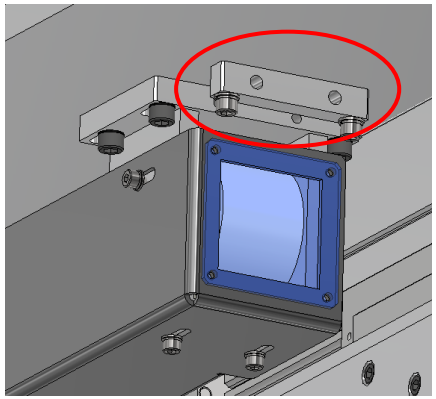
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

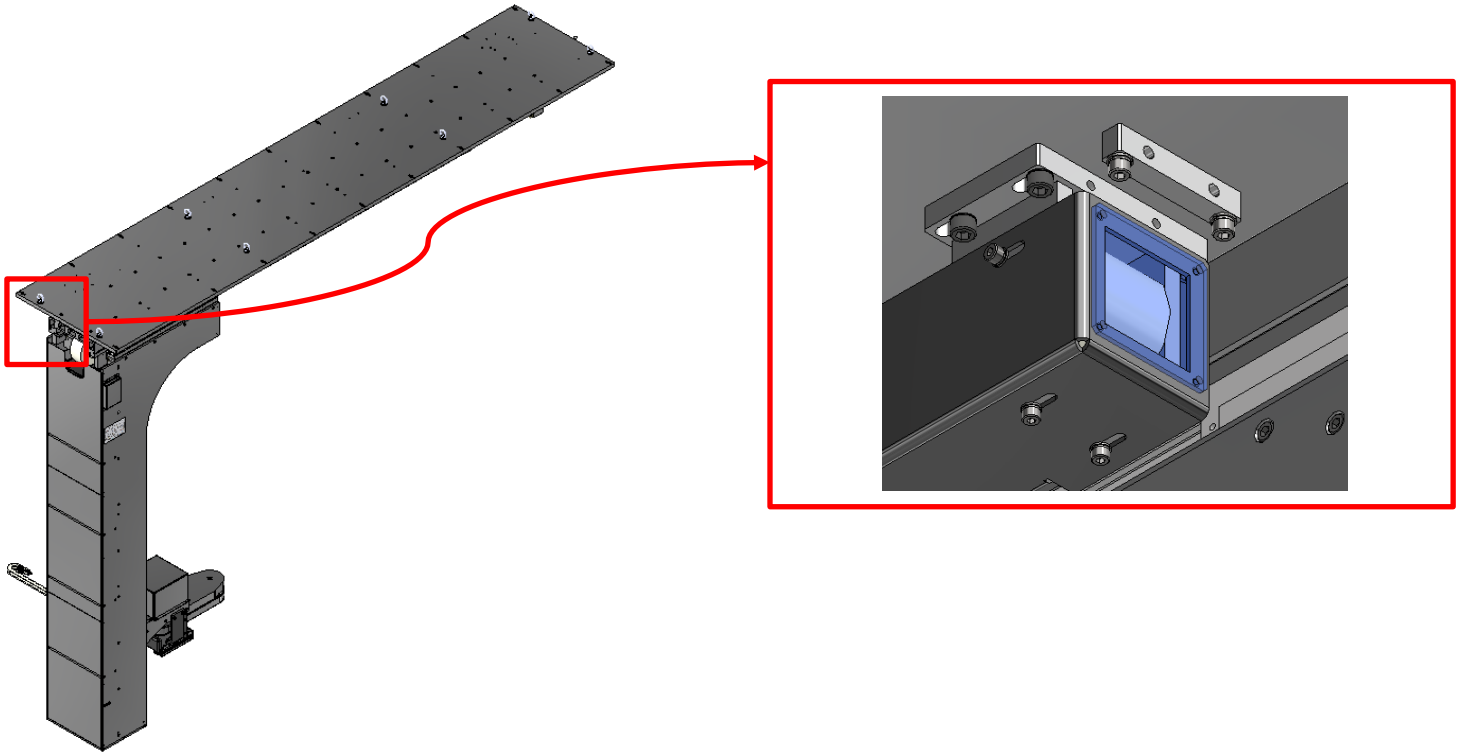
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #0	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO. ECN NO.	N/A ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Difficult to adjust the tilt due to a single tension adjustment hole on the Y-axis timing belt tension block.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
	4) FER Y AXIS PULLEY • 텐션조절볼트 1개로 ALIGN PULLEY 닿음 개선필요 		
상세내용 Detail	#0 P/N: TM01F-Y100-M006-0	Improved the tilt adjustment by modifying the Y-axis timing belt tension block to use 2 bolts. #0 P/N: TM01F-Y100-M006-1	

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전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

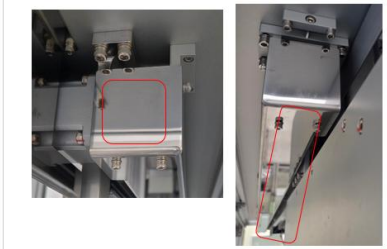
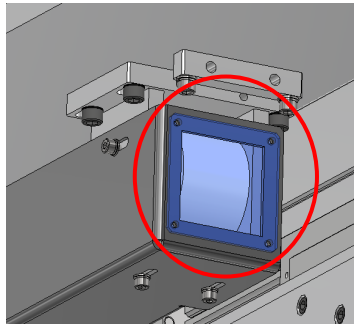
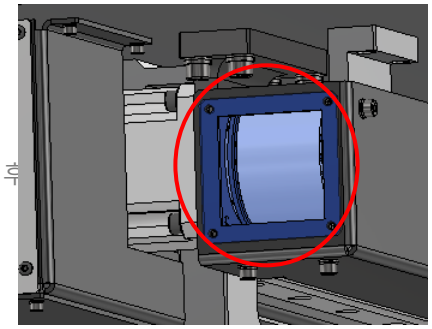
3. Attachments

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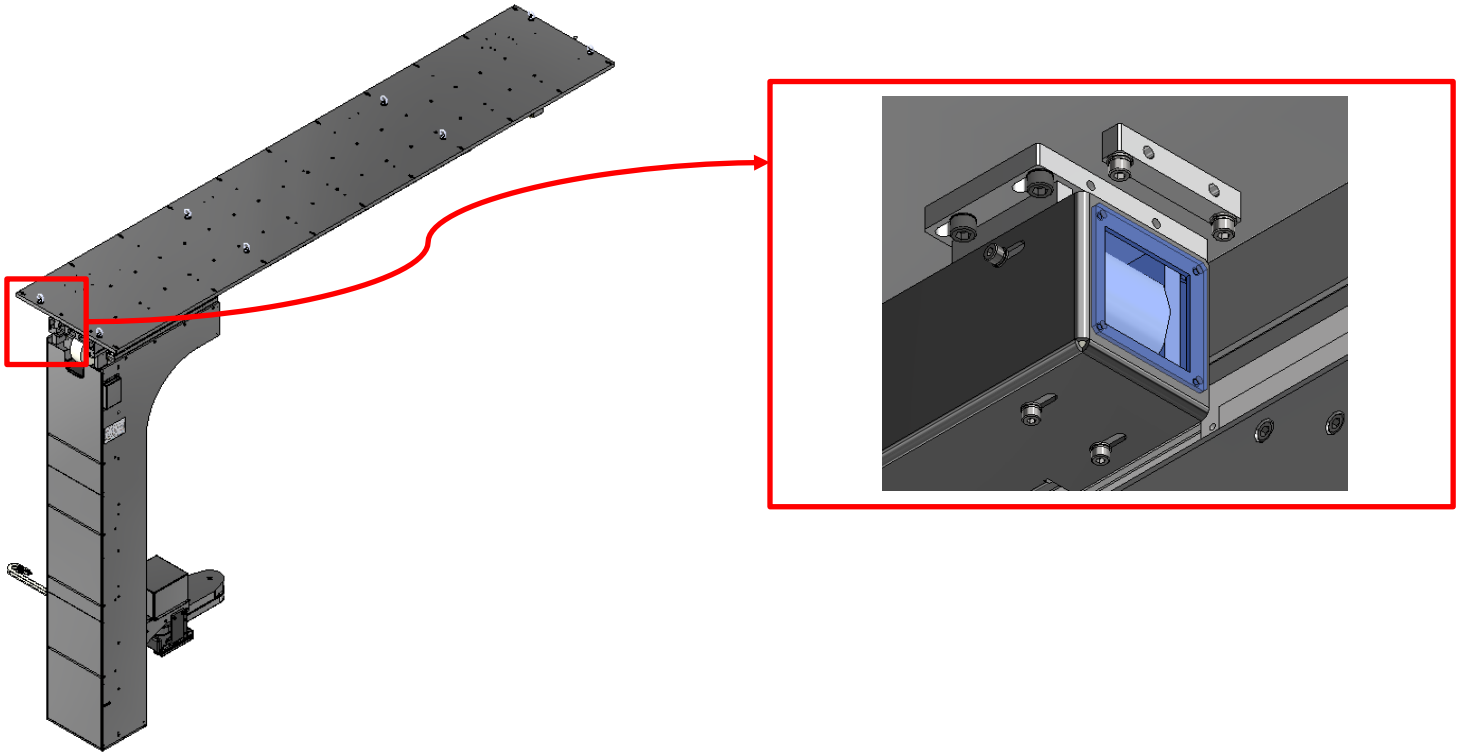
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Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #0	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO. ECN NO.	N/A ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Need to remove the cover to check the Y-axis timing belt misalignment.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
	5) FER Y AXIS PULLEY  • PULLEY 상태확인용으로 투명아크릴적용요청	 	
상세내용 Detail	#0 P/N: TM01F-Y100-P002-0, TM01F-Y100-P003-0	Added an inspection window to check the Y-axis timing belt misalignment. #0 P/N: TM01F-Y100-J001-0, TM01F-Y100-P002-1, TM01F-Y100-P003-1	

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유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

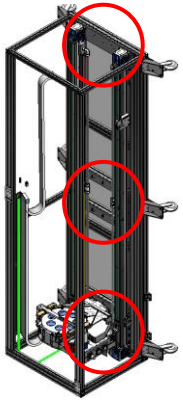
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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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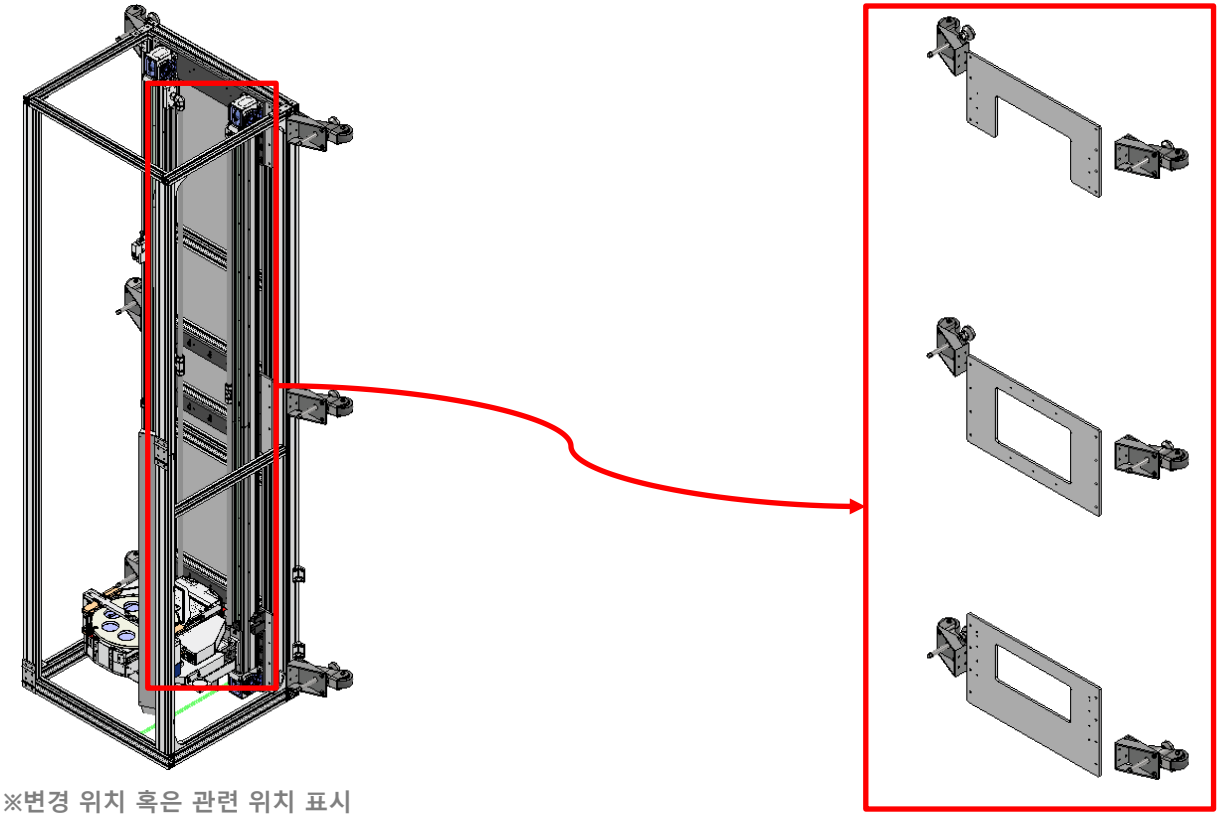
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-03-7813 IMMERSION TRACK PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #2,#3,#10,#11	적용 범위 Scope	■ 신규 발주분 (New order) ■ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Difficult to adjust the level with set screws due to the direct fastening between the shipping frame and the robot.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
	6) EMR, MPR, ESR CYCLE FRAME LEVEL  • 프로파일에 티칭무두볼트로 레벨 조절이 불가하여 FAT FRAME처럼 BASE에 안착되게 개선필요	
상세내용 Detail	#0 P/N: TM01F-Y100-P002-0, TM01F-Y100-P003-0	Added an intermediate plate between the shipping frame and the Z-axis base. #2,#3 P/N: TM01F-Y100-J001-0, TM01F-Y100-P002-1, TM01F-Y100-P003-1

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유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : / / 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

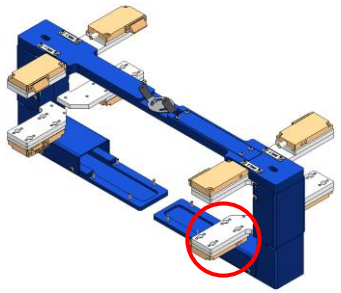
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
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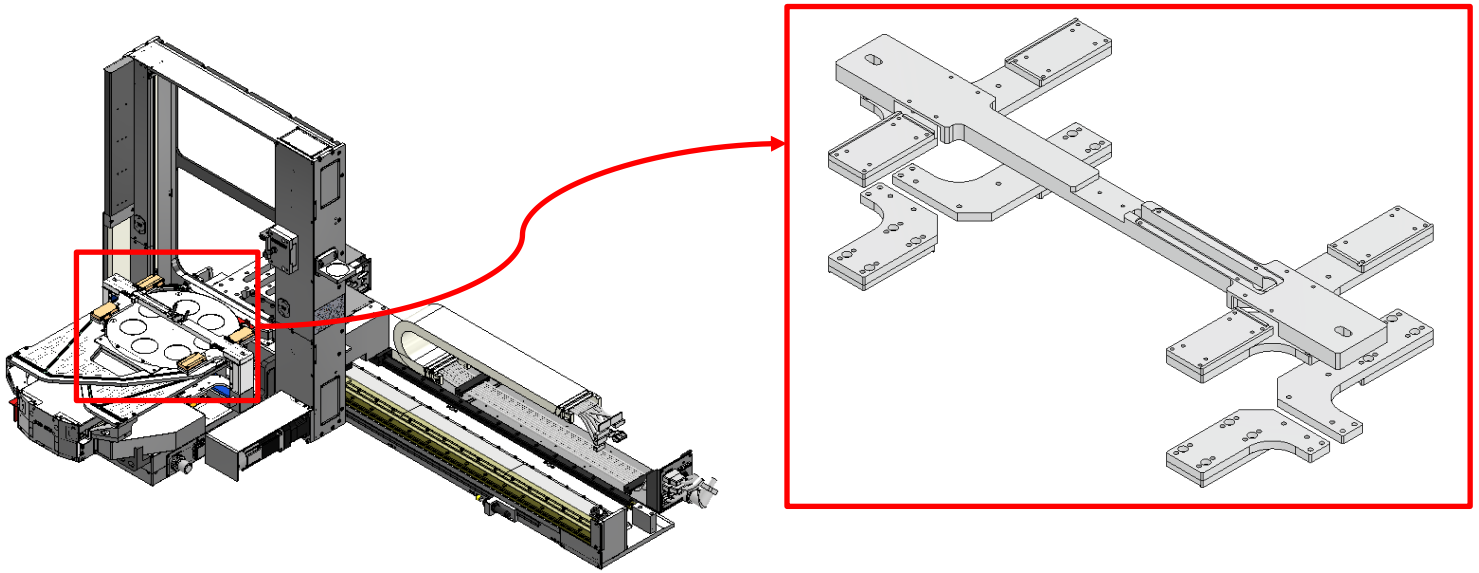
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-03-7813 IMMERSION TRACK PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1 ~ #12	적용 범위 Scope	■ 신규 발주분 (New order) ■ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Inconvenient to adjust the AWC sensor alignment.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
	8) AWC SETTING 취약 • 브라켓 고정이 단으로 밀착 조립이 되는 구조이지만, 볼트 체결 중 틀어짐이 쉽게 발생 • 외부 충격 등에 쉽게 변경됨 • 고정핀 또는 자리파기 등 개선이 필요함		
상세내용 Detail	Fine alignment required using shims between the AWC sensor alignment block and the sensor. #1 P/N: TM01E-X500-M001-0, TM01E-X500-M003-0, TM01E-X500-M003-0_M, TM01E-X500-M004-0, TM01E-X500-M004-0_M #2,#3,#10,#11 P/N: TM01S-X400-M005-0, TM01S-X400-M005-0_M, TM01S-X400-M009-0, TM01S-X400-M009-0_M, TM01S-X400-M010-0 #4,#5,#6,#7,#8,#9 P/N: TM01M-X500-M005-0, TM01M-X500-M005-0_M, TM01M-X500-M009-0, TM01M-X500-M009-0_M, TM01M-X500-M010-0 #12 P/N: TM01I-X500-M001-0, TM01I-X500-M003-0, TM01I-X500-M003-0_M, TM01I-X500-M004-0, TM01I-X500-M004-0_M	Added set screws to the block to allow direct alignment adjustment. #1 P/N: TM01E-X500-M001-1, TM01E-X500-M011-0, TM01E-X500-M011-0_M, TM01E-X500-M012-0, TM01E-X500-M012-0_M, TM01E-X500-M013-0, TM01E-X500-M013-0_M #2,#3,#10,#11 P/N: TM01S-X400-M010-1, TM01S-X400-M011-0, TM01S-X400-M011-0_M, TM01S-X400-M012-0, TM01S-X400-M012-0_M, TM01S-X400-M013-0, TM01S-X400-M013-0_M #4,#5,#6,#7,#8,#9 P/N: TM01M-X500-M010-1, TM01M-X500-M011-0, TM01M-X500-M011-0_M, TM01M-X500-M012-0, TM01M-X500-M012-0_M, TM01M-X500-M013-0, TM01M-X500-M013-0_M #12 P/N: TM01I-X500-M001-1, TM01I-X500-M011-0, TM01I-X500-M011-0_M, TM01I-X500-M012-0, TM01I-X500-M012-0_M, TM01I-X500-M013-0, TM01I-X500-M013-0_M	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

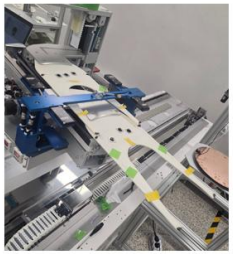
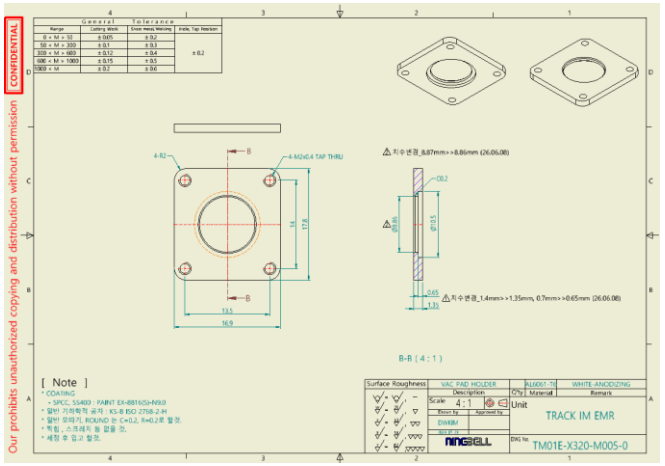
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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

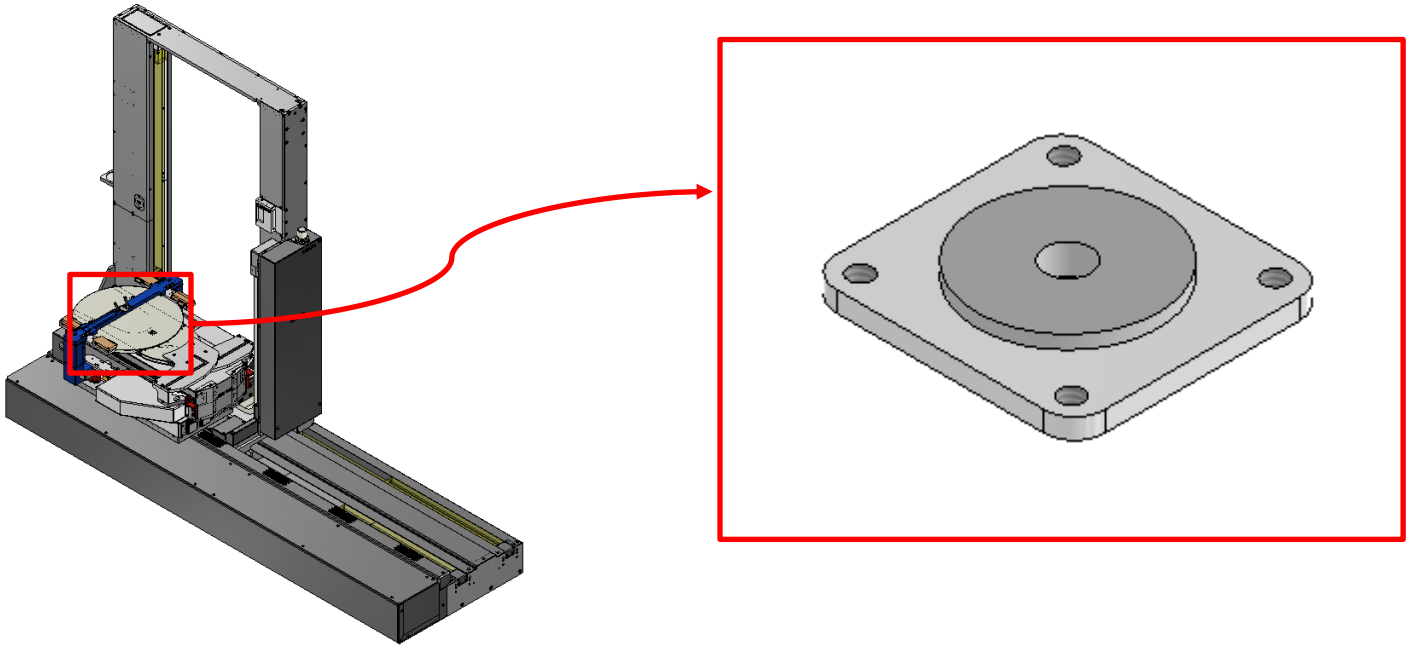
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-03-7813 IMMERSION TRACK PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1, #12	적용 범위 Scope	■ 신규 발주분 (New order) ■ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO. ECN NO.	N/A ECN-D-2607-02
변경 상세 사유 Detailed reason for change	A minor leak occurred at the O-ring between the ceramic end-effector and the arm.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
	9) EMR HAND 홀더리크 발생  • 세라믹핸드 오링부와 홀더부 미세리크로 개선필요	
상세내용 Detail	#1 P/N: TM01E-X320-M005-0 #12 P/N: TM01E-X320-M005-0	Machined the pad bracket step 0.05mm lower to compress the pad. #1 P/N: TM01E-X320-M005-1 #12 P/N: TM01E-X320-M005-1

2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

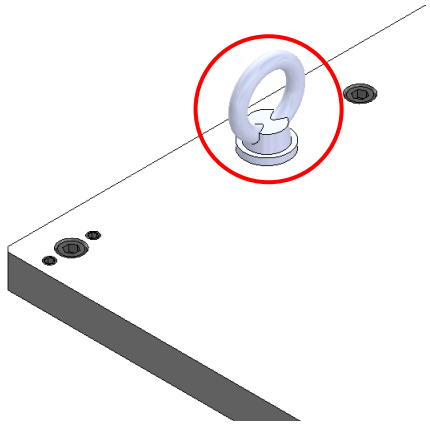
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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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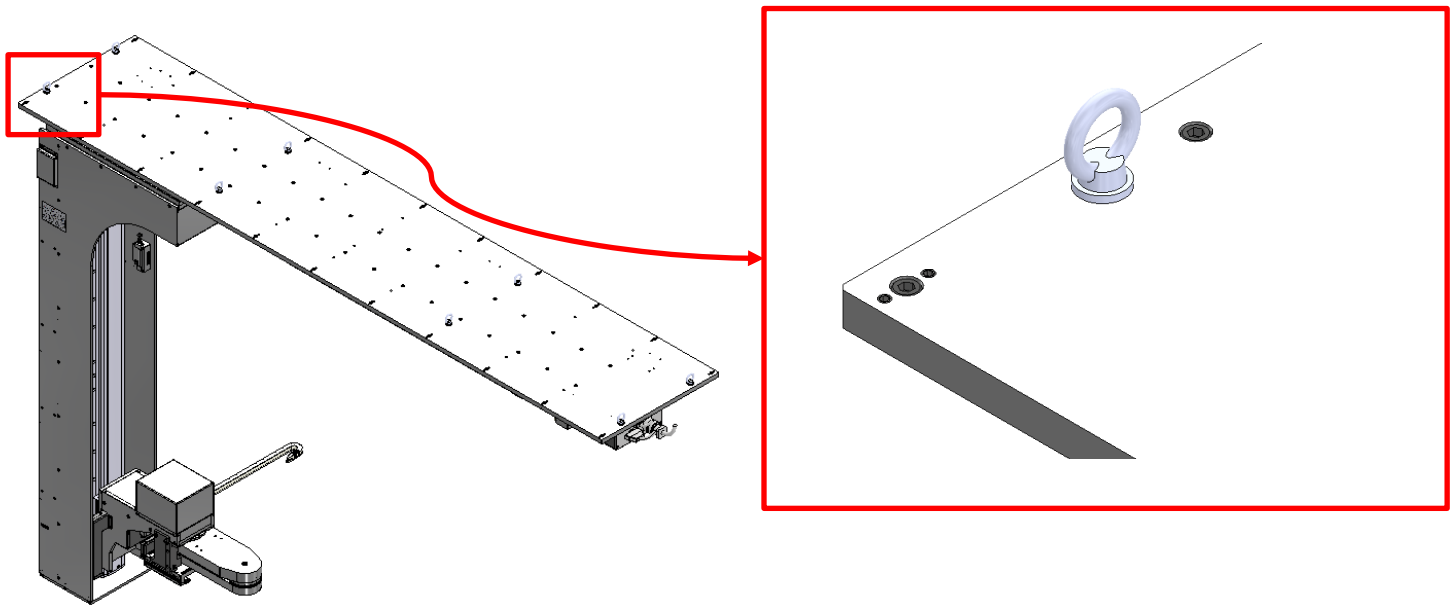
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #0	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO. ECN NO.	N/A ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Per ACM request, added eyebolts for the top of the FER to the E-BOM, along with urethane washers to prevent scratches.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
	10) 아이볼트 스크래치 개선필요 • 아이볼트 체결 시 Y축 베이스 스크래치 • 현장에서 로봇드는 체인 스크래치 발생 가능성 다수	
상세내용 Detail		Added eyebolts to the E-BOM. Added anti-scratch urethane washers. #0 P/N: J-01-660, J-07-113

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

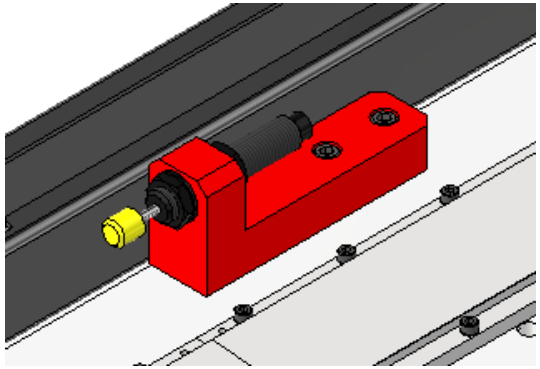
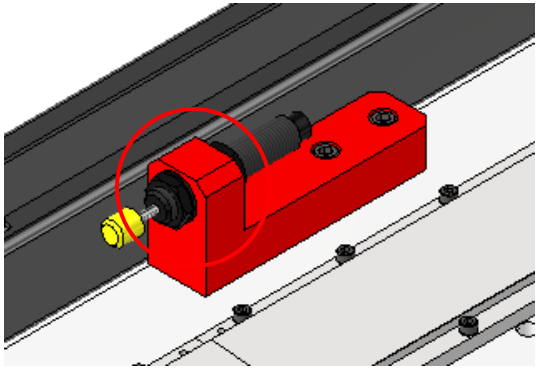
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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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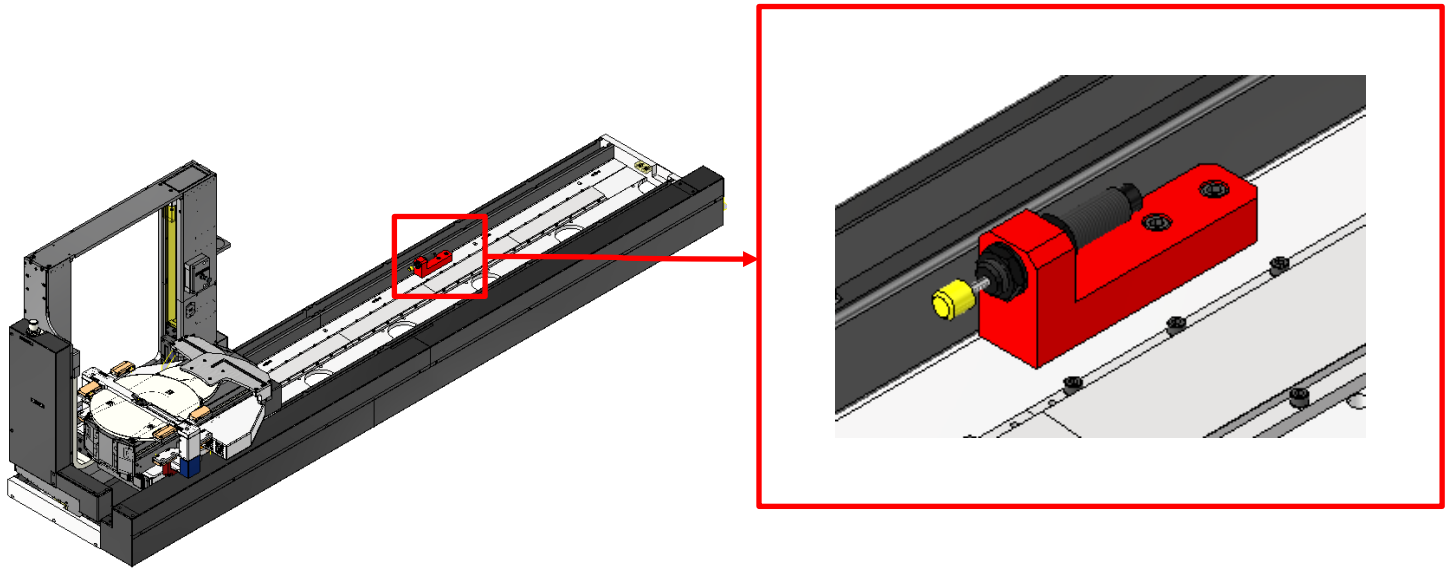
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1, #12	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Unable to fasten due to a thread pitch mismatch (Shock absorber: M14x1.5, Shipping block: M14x2.0).		

변경 사항 Changes	변경 전(Before)	변경 후(After)
		
상세내용 Detail	#1 P/N: TM01E-Y100-M020-0, TM01E-Y100-M020-0_M #12 P/N: TM01I-Y100-M014-0, TM01I-Y100-M014-0_M	Changed the shipping block thread pitch to M14x1.5. #1 P/N: TM01E-Y100-M020-1, TM01E-Y100-M020-1_M #12 P/N: TM01I-Y100-M014-1, TM01I-Y100-M014-1_M

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

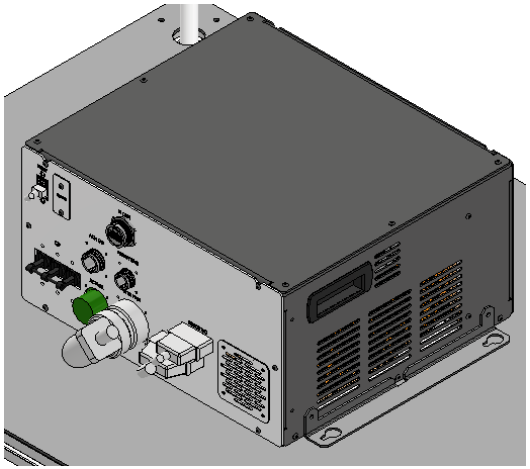
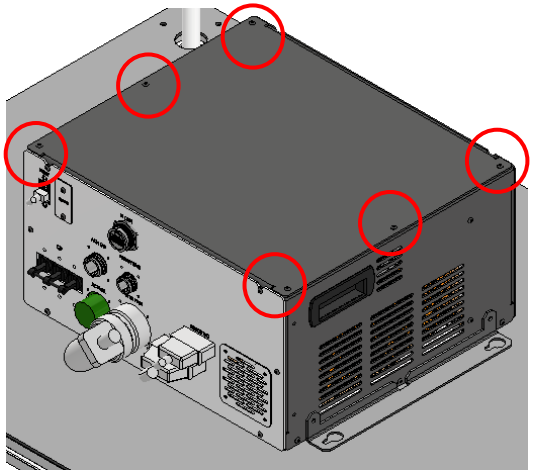
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
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2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

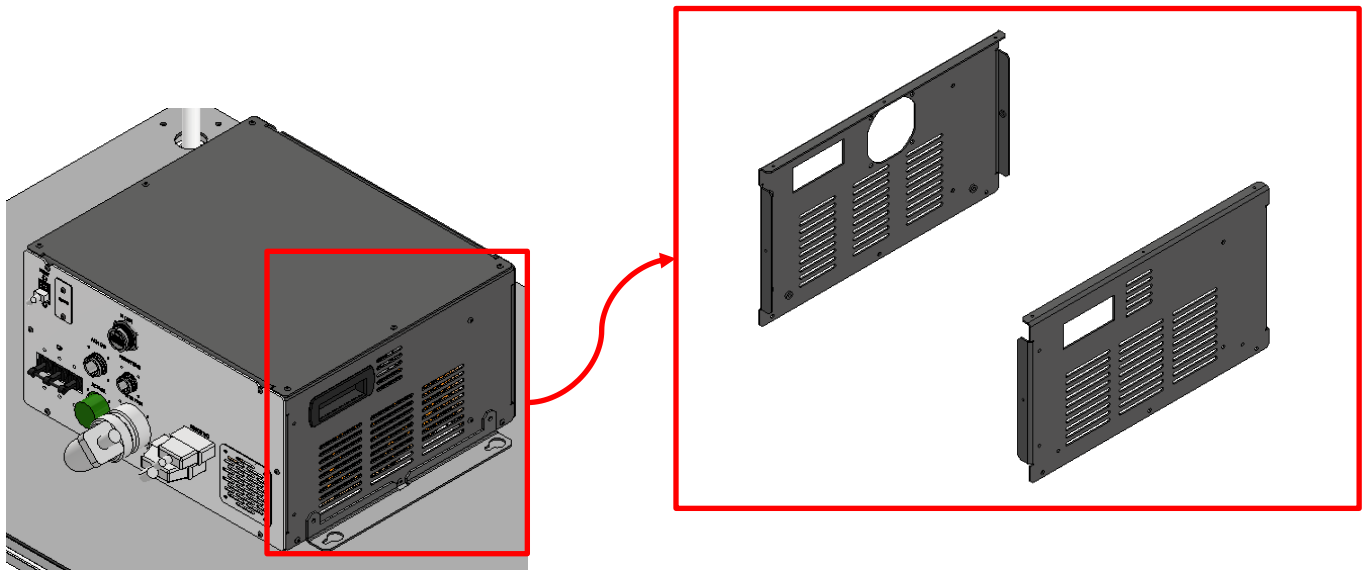
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #0 ~ #12	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Intermittent thread stripping when fastening the control box top cover.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail	#0 ~ #12 P/N: TM01C-C100-P006-1, TM01C-C100-P005-1	Changed from an M3 tapped hole on the 1.5T mating cover to a clinching nut inserted from the opposite side. #0 ~ #12 P/N: TM01C-C100-P006-2, TM01C-C100-P005-2	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

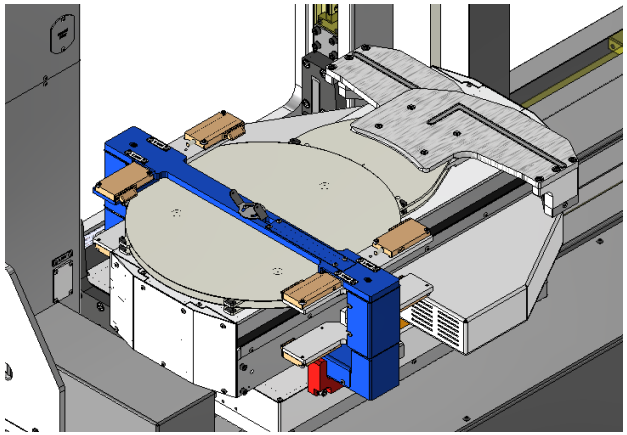
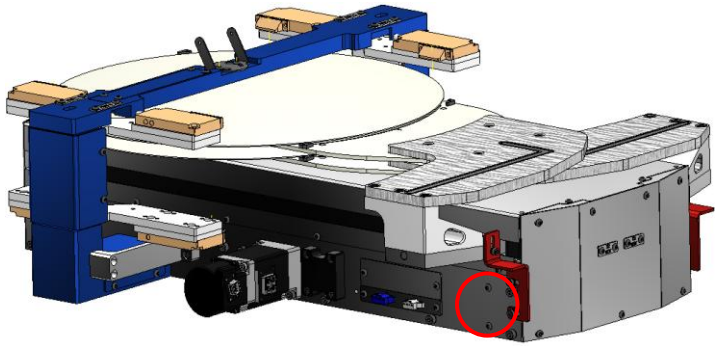
3. Attachments

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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

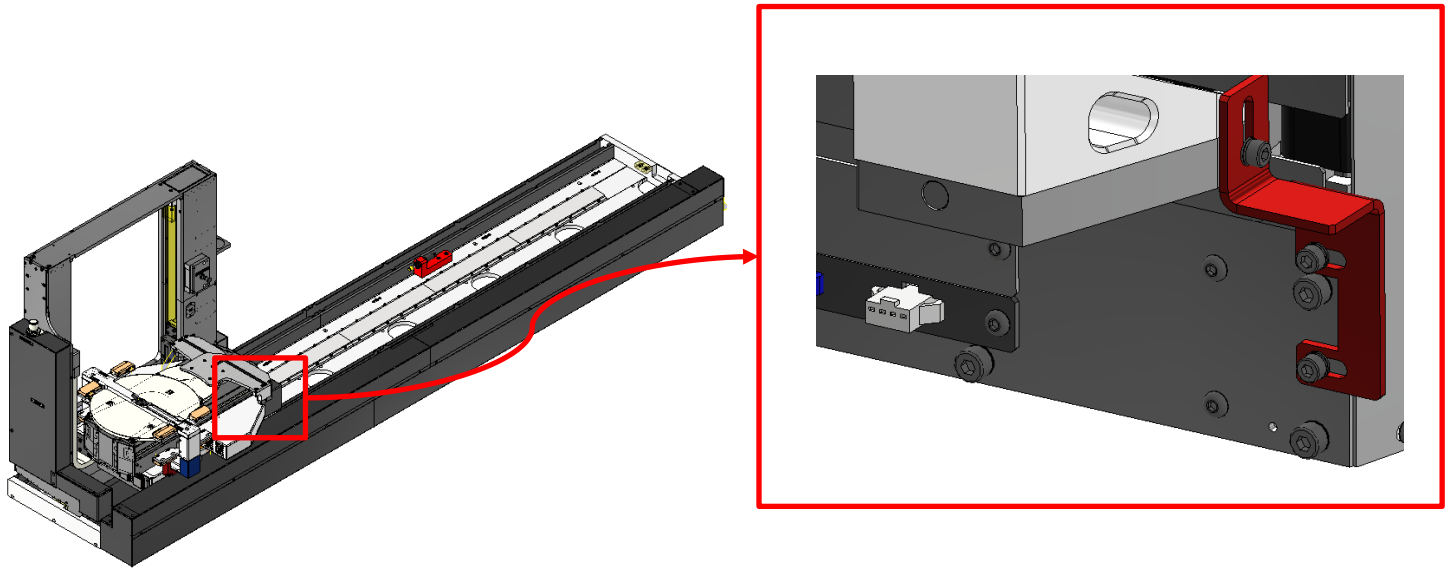
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1, #12	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Removed unused holes in EMR and ITR.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail	#1 P/N: TM01E-X600-M002-1, TM01E-X600-M002-1_M #12 P/N: TM01I-X600-M002-1, TM01I-X600-M002-1_M	Removed the idler mounting hole. #1 P/N: TM01E-X600-M002-2, TM01E-X600-M002-2_M #12 P/N: TM01I-X600-M002-2, TM01I-X600-M002-2_M	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

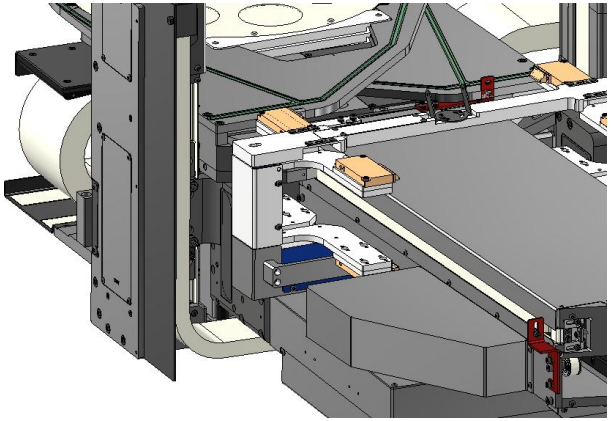
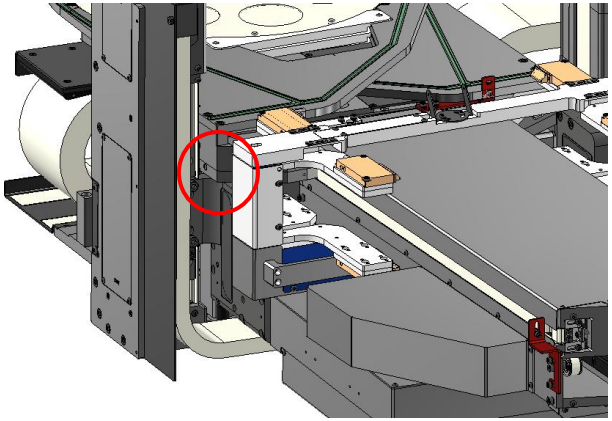
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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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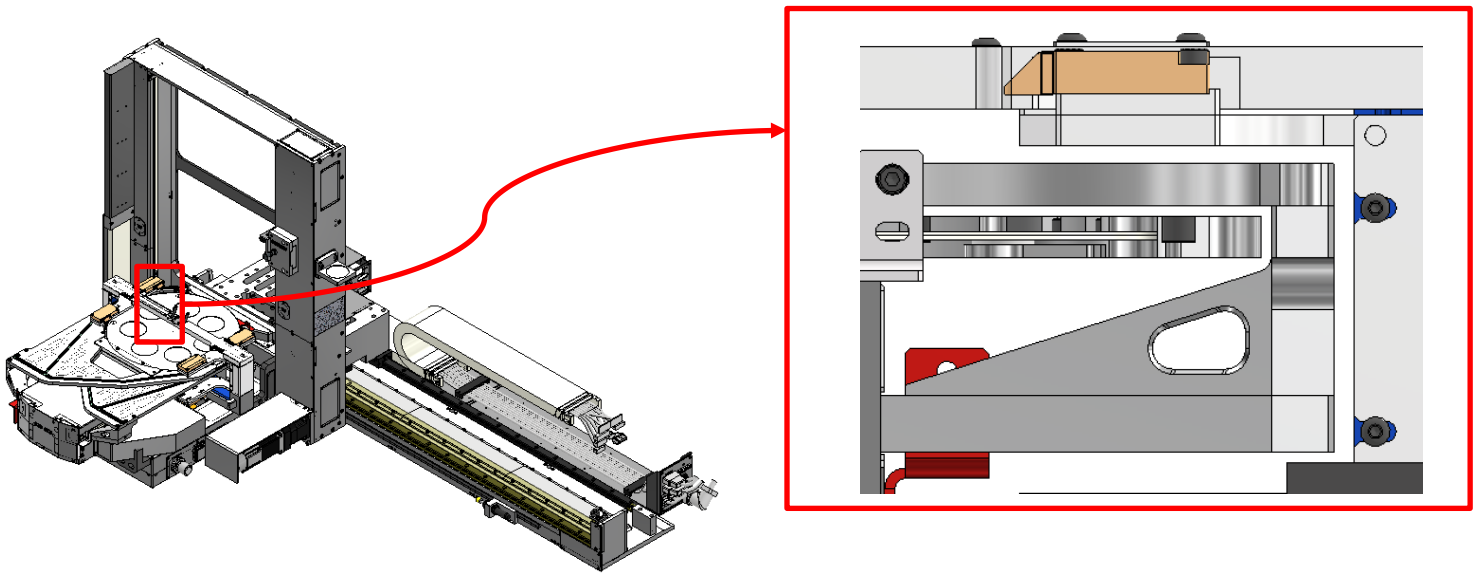
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #4 ~#9	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	1mm gap clearance with the Z-axis mover when using the maximum X-axis stroke at 2.7deg in the Bake direction.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
		
상세내용 Detail	#4,#5,#6 P/N: TM01M-Z100-M019-0 #7,#8,#9 P/N: TM01M-Z100-M019-0_M	Modified the Z-axis mover to increase the gap: 1mm -> 2mm. #4,#5,#6 P/N: TM01M-Z100-M019-1 #7,#8,#9 P/N: TM01M-Z100-M019-1_M

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

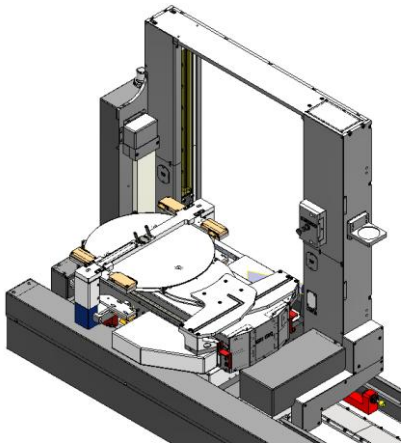
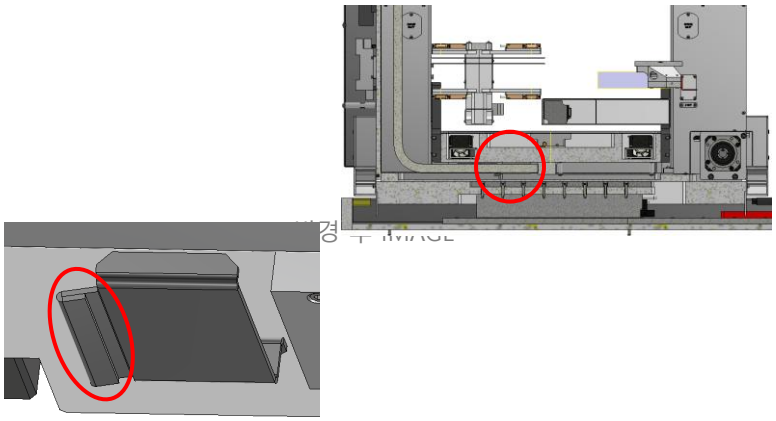
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N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

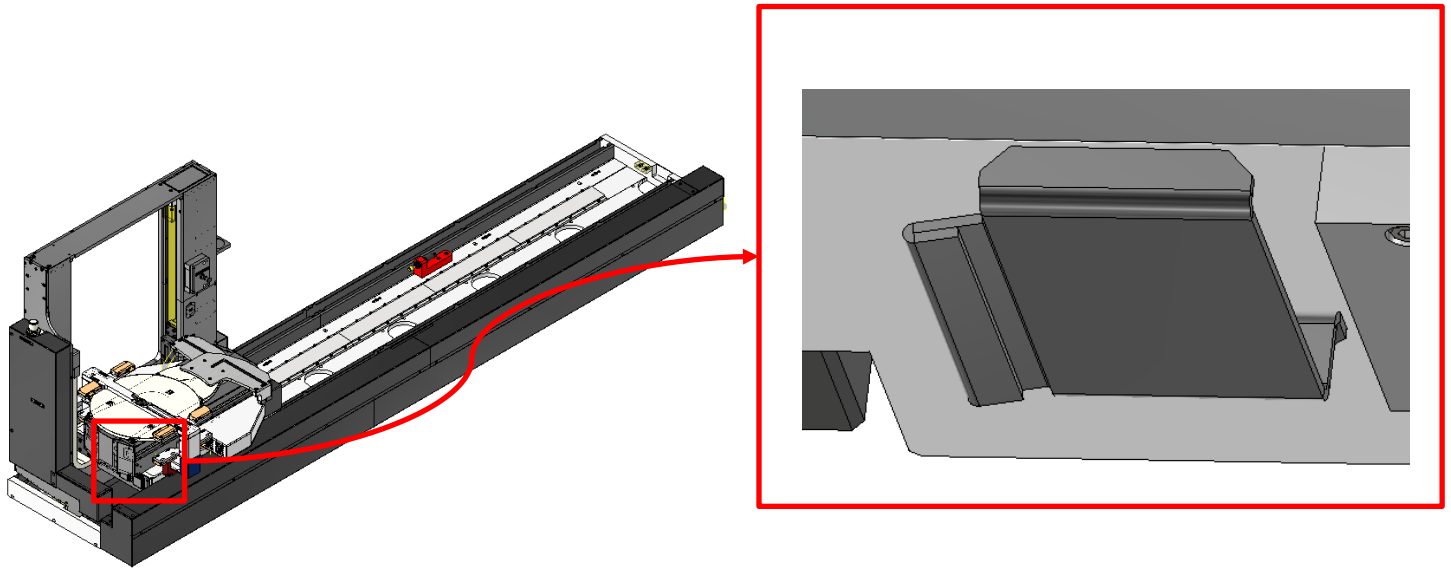
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-03-7813 IMMERSION TRACK PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1, #12	적용 범위 Scope	■ 신규 발주분 (New order) ■ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	The trackless cable is rubbing against the edge of the Z-axis trackless cable SUS bracket.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail	#1 P/N: TM01E-T100-P007-1 #12 P/N: TM01I-T100-P007-1	Hemmed the edge of the trackless cable SUS bracket. #1 P/N: TM01E-T100-P007-1 #12 P/N: TM01I-T100-P007-1	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

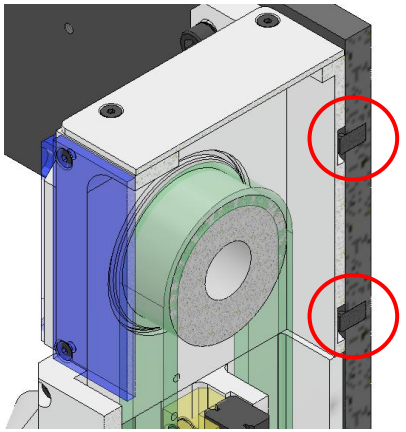
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

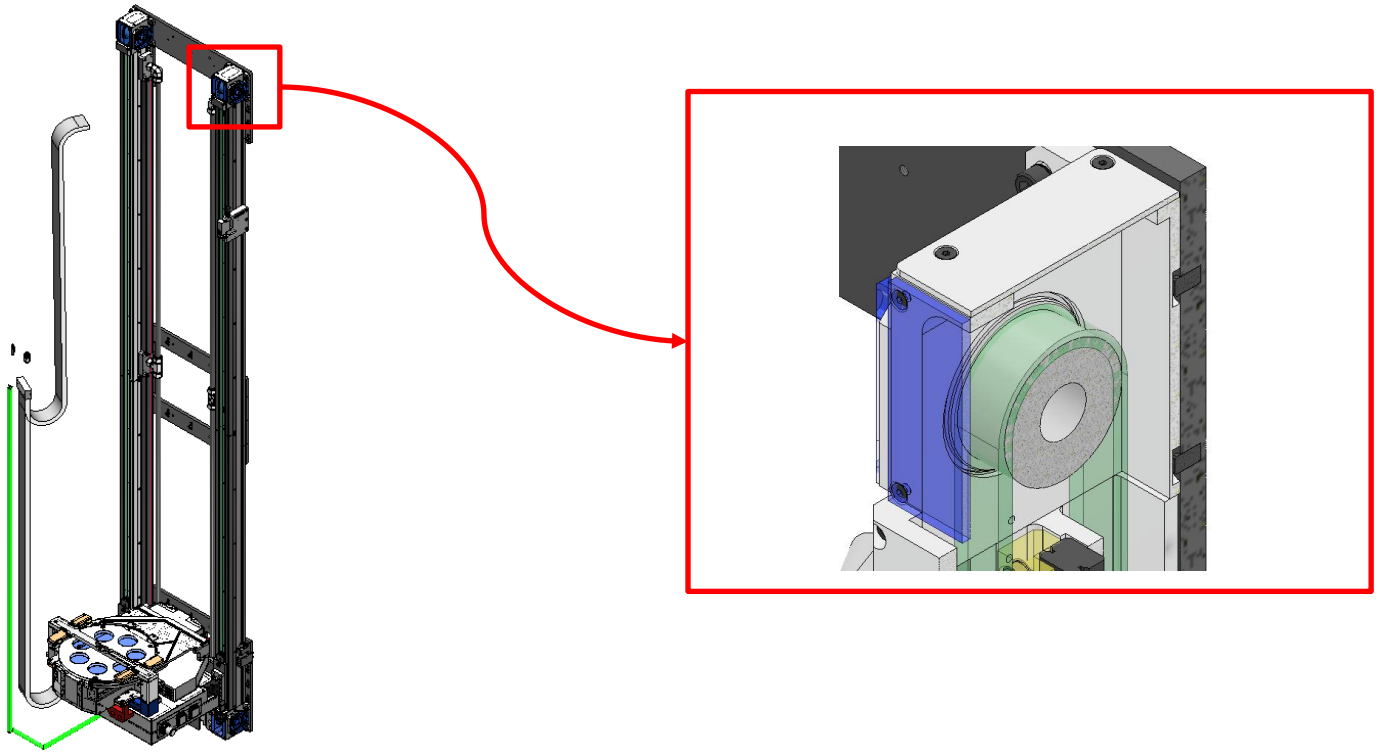
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #2, #3, #10, #11	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	The Z-axis pulley box pin hole is a through-hole.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
	변경 전 IMAGE		
상세내용 Detail	#2,#3 P/N: TM01S-Z110-M003-1, TM01S-Z130-M001-0 #10,#11 P/N: TM01S-Z110-M003-1, TM01S-Z130-M001-0	Changed the Z-axis pulley box pin hole to a blind hole. #2,#3 P/N: TM01S-Z110-M003-2, TM01S-Z130-M001-1 #10,#11 P/N: TM01S-Z110-M003-2, TM01S-Z130-M001-1	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / N	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	Y / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

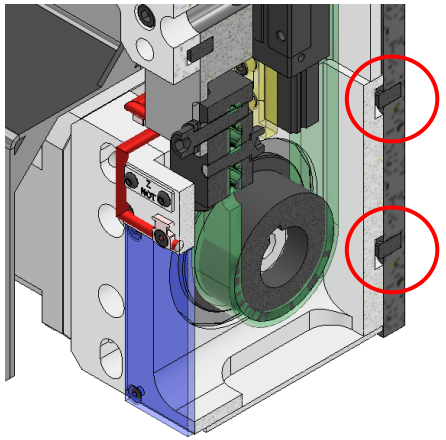
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

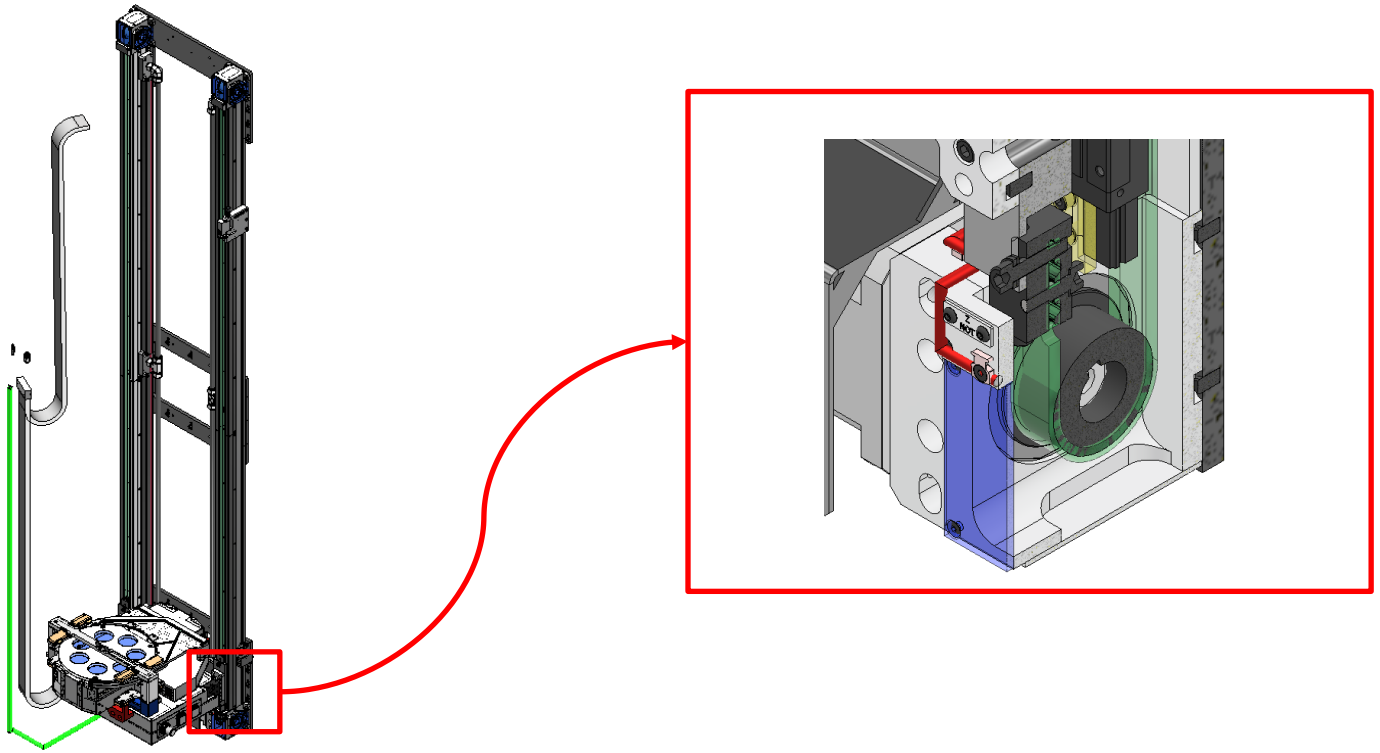
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #2, #3, #10, #11	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	The Z-axis motor pulley box pin hole is too shallow. Required using a 6x8 dowel pin instead of the standard 6x10 dowel pin.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
	변경 전 IMAGE		
상세내용 Detail	#2,#3 P/N: TM01S-Z110-M001-1, TM01S-Z110-M001-1_M #10,#11 P/N: TM01S-Z110-M001-1, TM01S-Z110-M001-1_M	Adjusted the Z-axis motor pulley box pin hole depth. #2,#3 P/N: TM01S-Z110-M001-1, TM01S-Z110-M001-1_M #10,#11 P/N: TM01S-Z110-M001-1, TM01S-Z110-M001-1_M	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

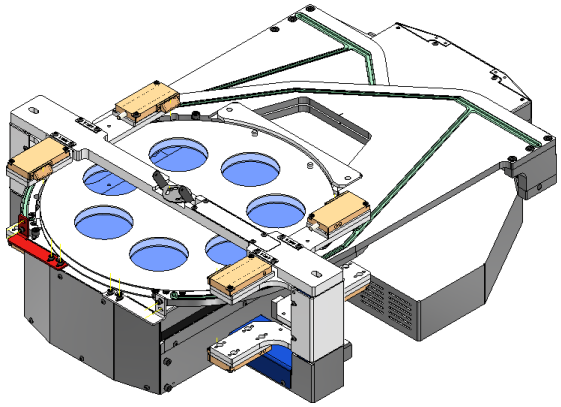
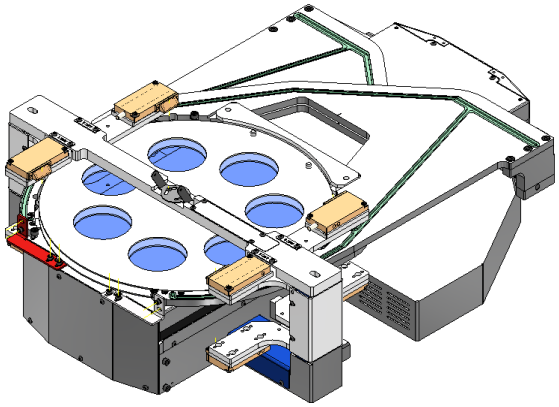
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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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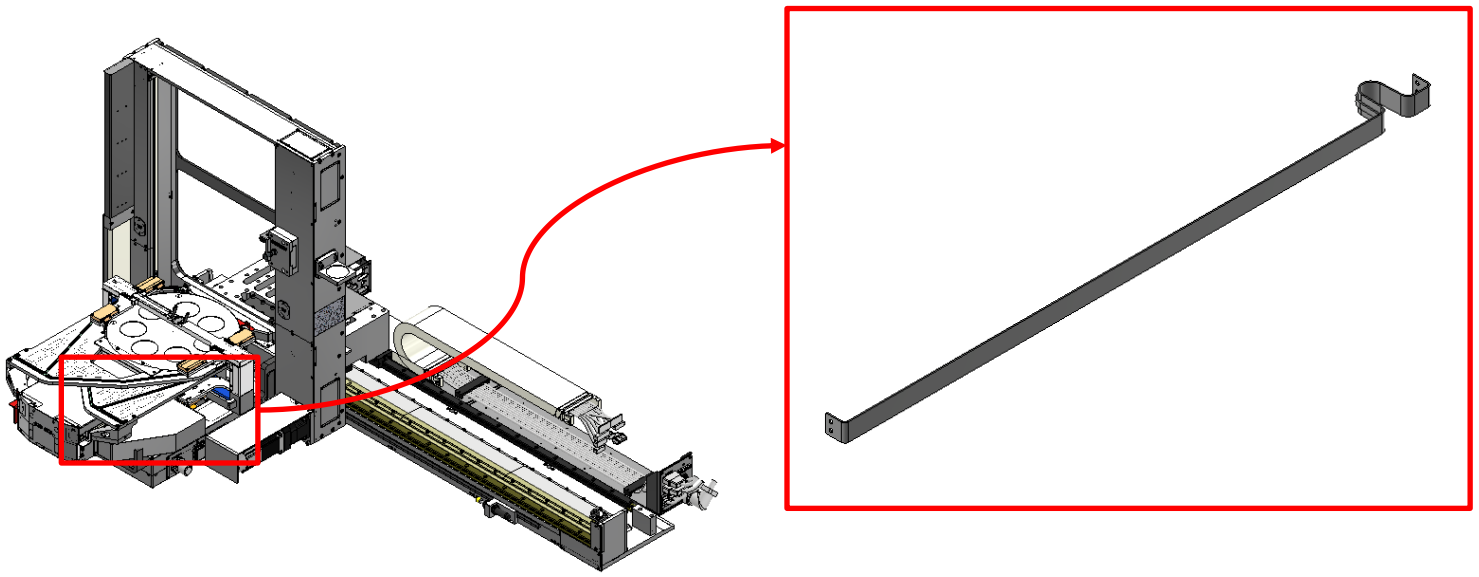
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-03-7813 IMMERSION TRACK PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1 ~ #12	적용 범위 Scope	■ 신규 발주분 (New order) ■ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	X-axis sealing belt breakage occurred on some robots.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
		
상세내용 Detail	#1 P/N: TM01E-BT01-M004-2 #2,#3,#10,#11 P/N: TM01S-BT01-M006-2 #4,#5,#6,#7,#8,#9 P/N: TM01M-BT01-M002-2 #12 P/N: TM01I-BT01-M004-2	Changed the sealing belt tension cord: KEVLAR -> GLASS FIBER. Currently conducting marathon testing and monitoring of the GLASS FIBER in the 5th-floor lab. #1 P/N: TM01E-BT01-M004-3 #2,#3,#10,#11 P/N: TM01S-BT01-M006-3 #4,#5,#6,#7,#8,#9 P/N: TM01M-BT01-M002-3 #12 P/N: TM01I-BT01-M004-3

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

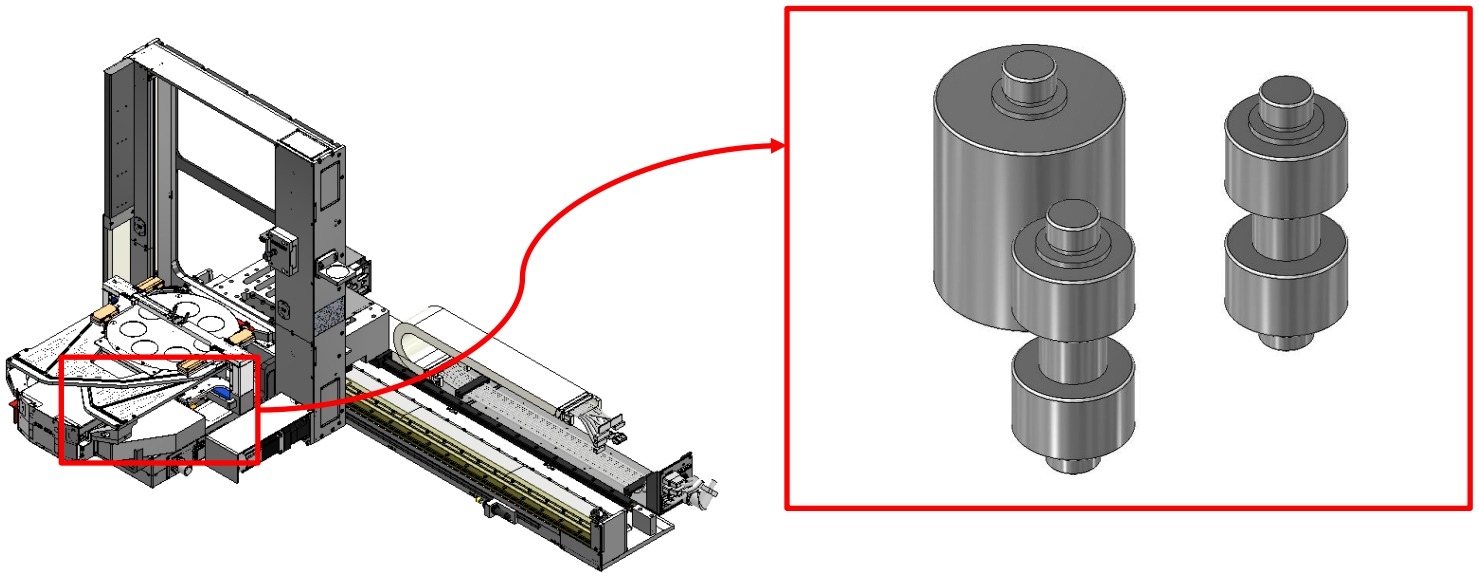
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1 ~ #12	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Difficult to machine the minimum corner radius (Min. R) on the X-axis sealing belt idler.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
	변경 전 IMAGE		
상세내용 Detail	#1 P/N: TM01E-X600-M060-0, TM01E-X600-M061-0 #2,#3,#10,#11 P/N: TM01S-X100-M028-0, TM01S-X100-M029-0 #4,#5,#6,#7,#8,#9 P/N: TM01M-X100-M023-0, TM01M-X100-M031-0 #12 P/N: TM01I-X600-M060-0, TM01I-X600-M061-0	Improved manufacturability by adding an undercut. #1 P/N: TM01E-X600-M060-1, TM01E-X600-M061-1 #2,#3,#10,#11 P/N: TM01S-X100-M028-1, TM01S-X100-M029-1 #4,#5,#6,#7,#8,#9 P/N: TM01M-X100-M023-1, TM01M-X100-M031-1 #12 P/N: TM01I-X600-M060-1, TM01I-X600-M061-1	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

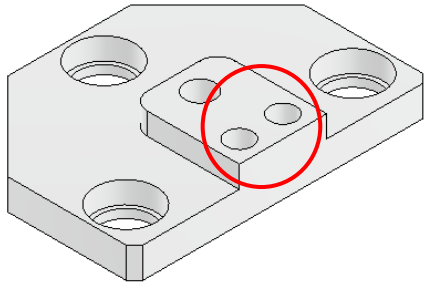
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Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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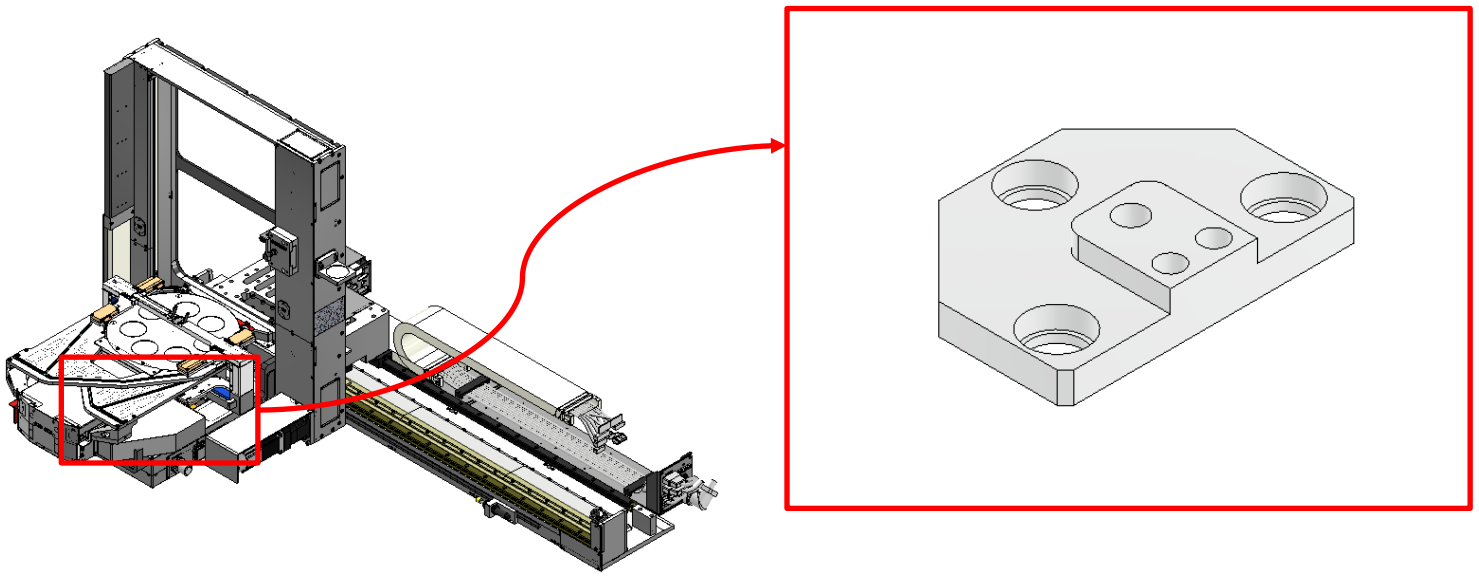
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1 ~ #12	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	The X-axis sealing belt block rotates due to fastening torque. The design requires careful alignment by the operator during assembly.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
	변경 전 IMAGE	
상세내용 Detail	#1 P/N: TM01E-X600-M042-0 #2,#3,#10,#11 P/N: TM01S-X100-M026-0 #4,#5,#6,#7,#8,#9 P/N: TM01M-X100-M016-0 #12 P/N: TM01I-X600-M042-0	Added 2 pins to the X-axis sealing belt block to prevent rotation. Note: 3x8 dowel pins must be used due to depth limitations. #1 P/N: TM01E-X600-M042-1 #2,#3,#10,#11 P/N: TM01S-X100-M026-1 #4,#5,#6,#7,#8,#9 P/N: TM01M-X100-M016-1 #12 P/N: TM01I-X600-M042-1

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

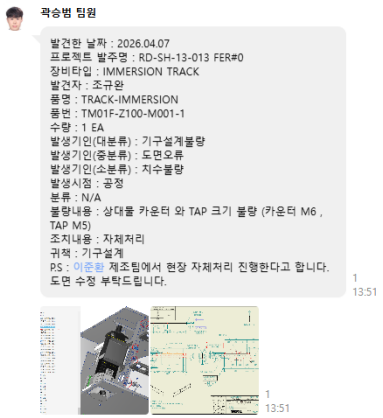
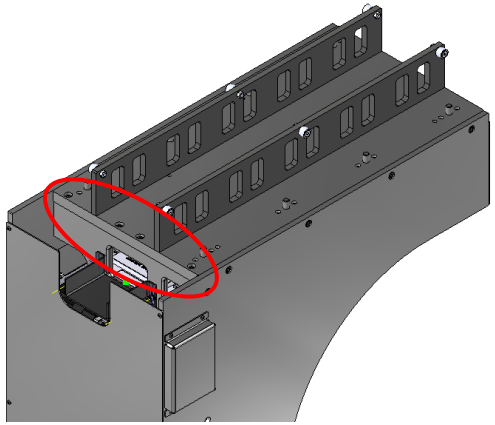
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

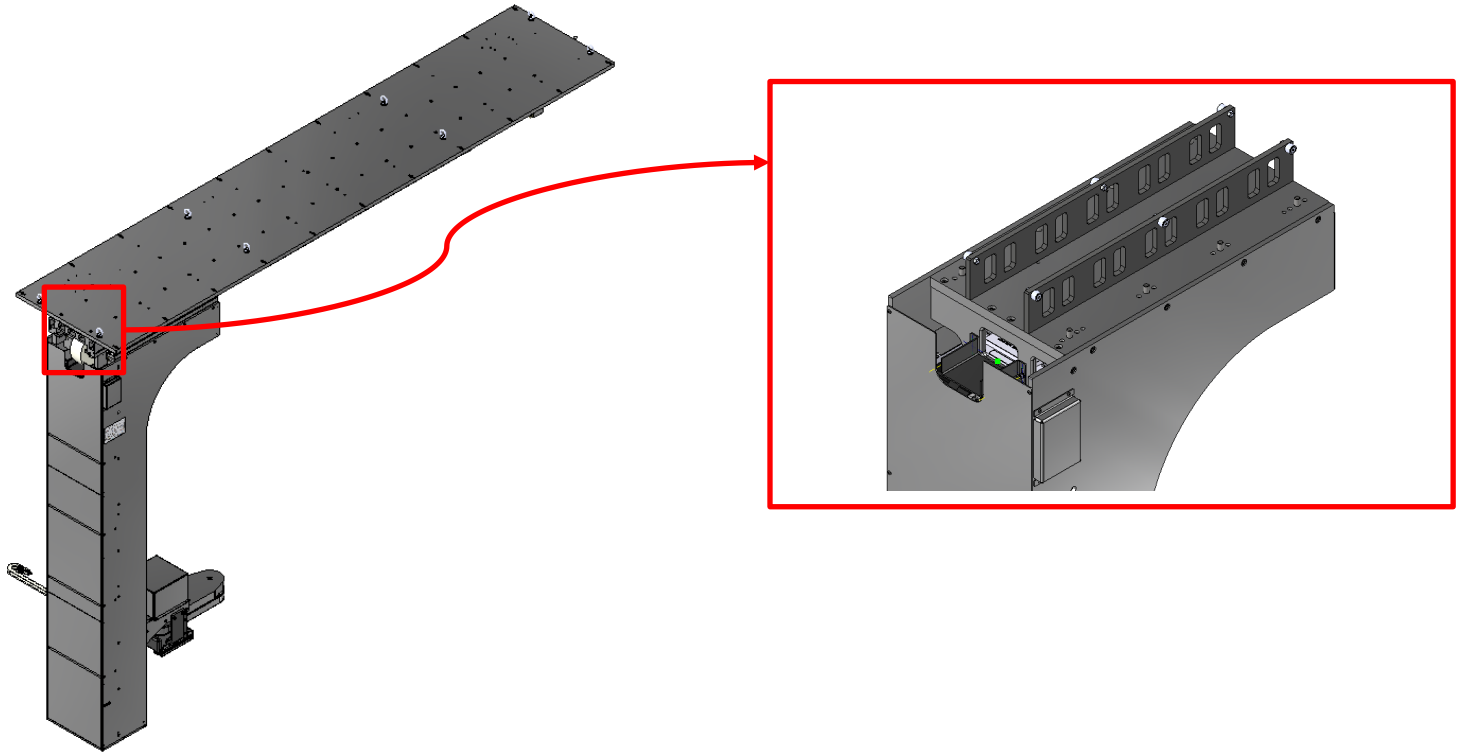
CONFIDENTIAL

2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #0	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Mating part counterbore and tap size mismatch (Counterbore: M6, Tap: M5).		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail	#0 P/N: TM01F-Z100-M001-0	Changed the mating part tap: M5 -> M6. #0 P/N: TM01F-Z100-M001-1	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

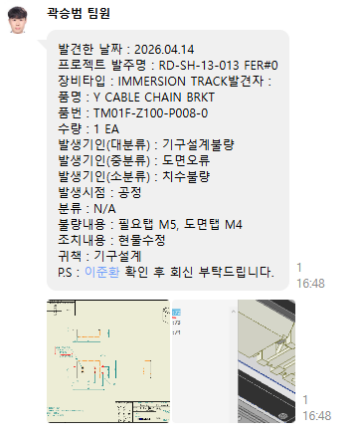
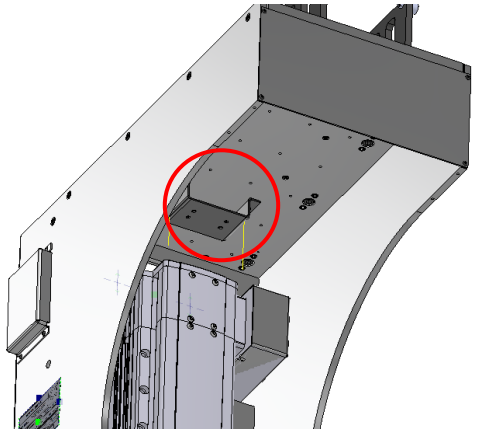
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

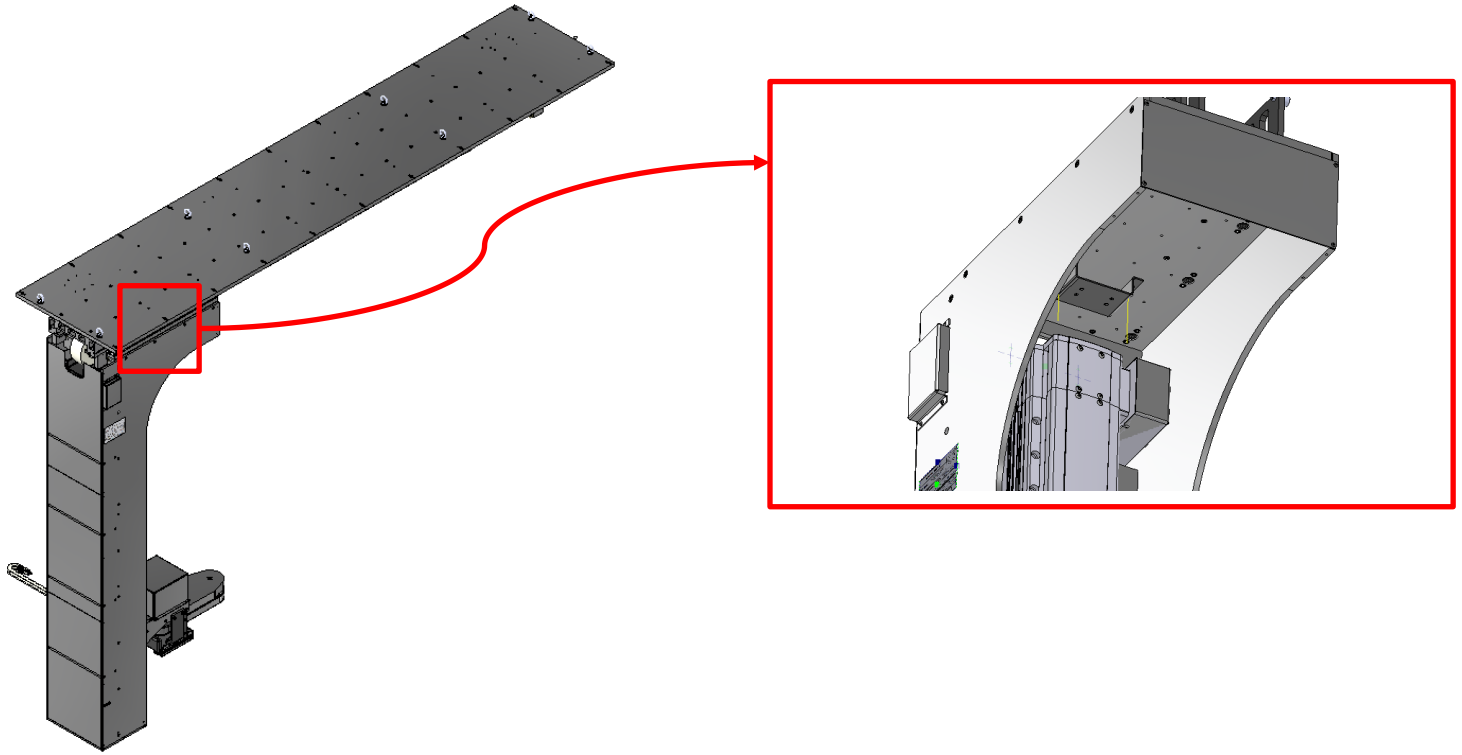
CONFIDENTIAL

2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #0	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO. ECN NO.	N/A ECN-D-2607-02
변경 상세 사유 Detailed reason for change	The required tap is M5, but it is specified as M4 on the drawing.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail	#0 P/N: TM01F-Z100-M004-0	Changed the mating part tap: M4 -> M5. #0 P/N: TM01F-Z100-M004-1	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

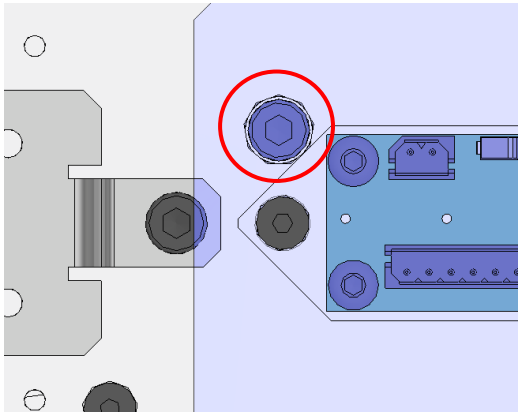
N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

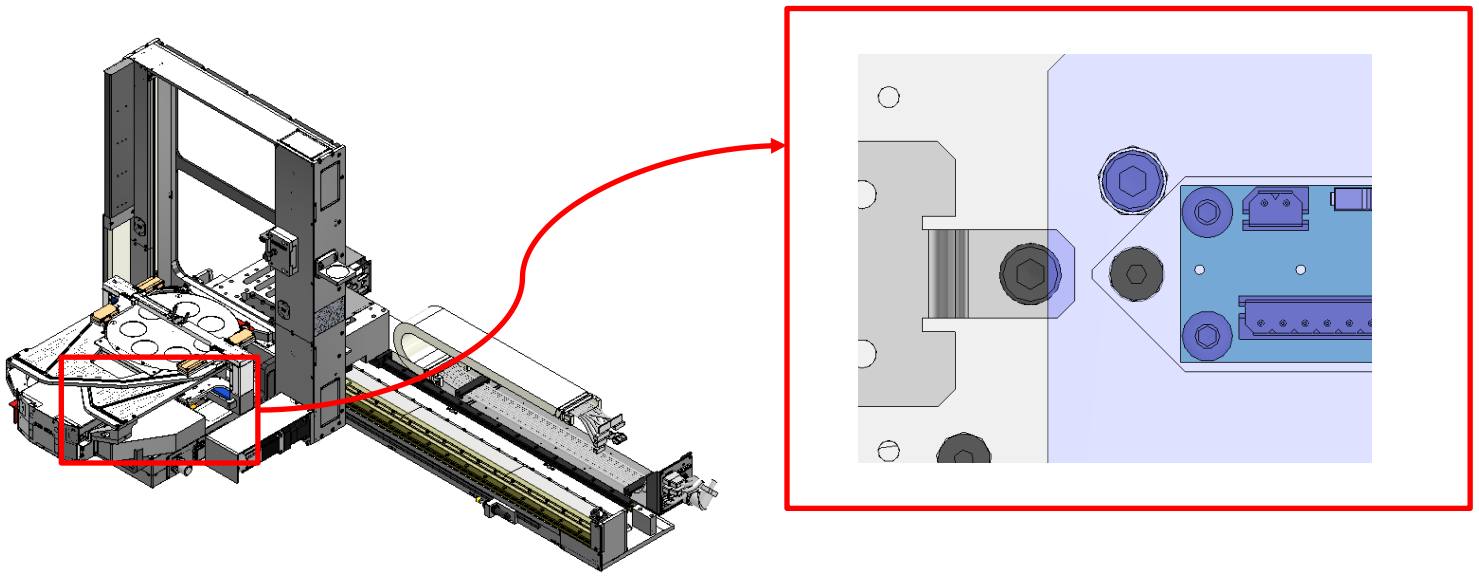
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1 ~ #12	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Need to remove the I/O board bracket when assembling or disassembling the X-axis base.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
	발견한 날짜 : 2026.04.16 프로젝트 발주명 : RD-SH-13-013 MPR #4, #5, #6 장비타입 : IMMERSION TRACK 발견자 : 김보석 품명 : 1) X IO BOARD BRKT 2) X MIDDLE COVER 품번 : 1) TM01M-X100-M022-0 2) TM01M-X100-M020-0 수량 : 각 3 EA (총 6EA) 발생기인(대분류) : 기구설계불량 발생기인(중분류) : 도면오류 발생기인(소분류) : 간섭불량 발생시점 : 공정 분류 : N/A 불량내용 : 보드 브라켓과 미들 커버 간섭으로 볼트 체결불가 조치내용 : 자체처리 귀책 : 기구설계 P.S : 금번은 제조팀 배선 후 재조립으로 자체처리 예정입니다. 자기 프로젝트에서는 동일 사항 발생하지 않도록 조치 방안 회신 부탁드리겠습니다. 이준환 1 11:21  1 11:21	
상세내용 Detail	#1 P/N: TM01E-X600-M022-0 #2,#3,#10,#11 P/N: TM01S-X100-M019-0 #4,#5,#6,#7,#8,#9 P/N: TM01M-X100-M022-0 #12 P/N: TM01I-X600-M022-0	Modified the I/O board bracket shape to avoid the DD motor mounting bolt holes. #1 P/N: TM01E-X600-M022-1 #2,#3,#10,#11 P/N: TM01S-X100-M019-1 #4,#5,#6,#7,#8,#9 P/N: TM01M-X100-M022-1 #12 P/N: TM01I-X600-M022-1

2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

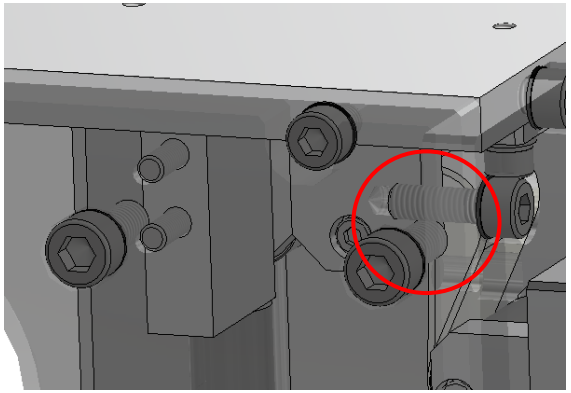
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

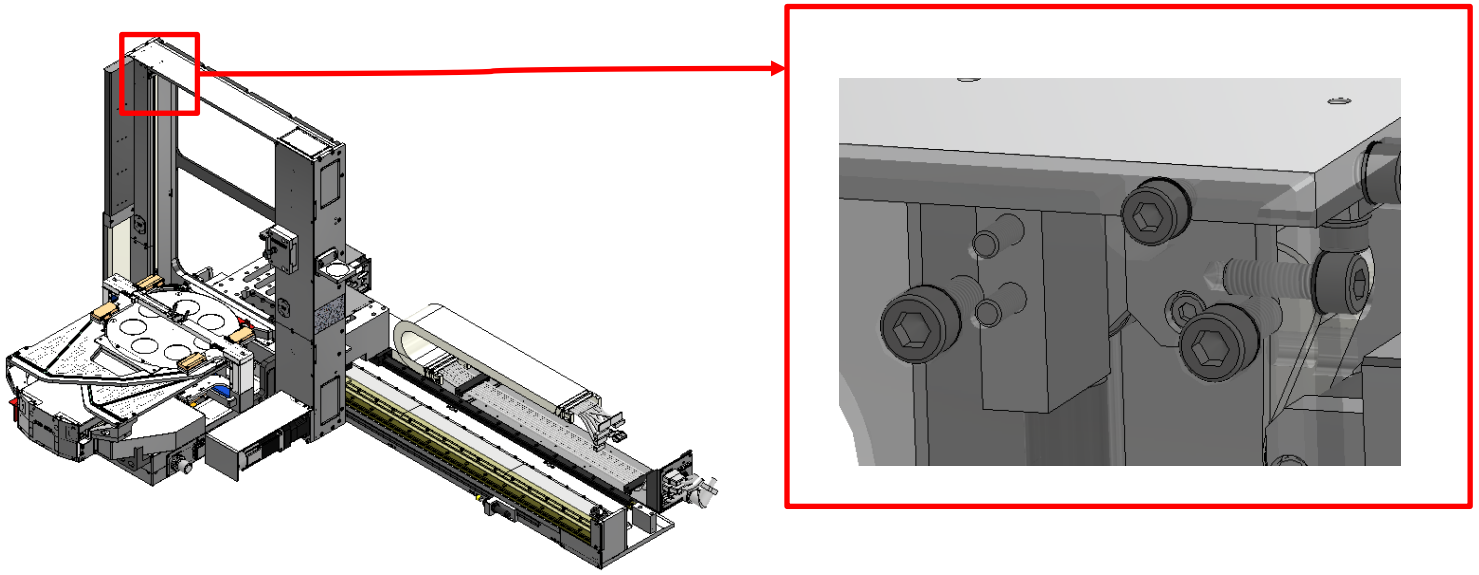
CONFIDENTIAL

2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #4 ~ #9	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Interference between the back plate and side plate mounting bolt holes during Z-axis base assembly.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
	곽승범 팀원 발견한 날짜 : 2026.04.17 프로젝트 발주명 : RD-SH-13-013 MPR #4, #5, #6 장비타입 : IMMERSION TRACK 발견자 : 길보석 품명 : Z REINFORCED BACK PLATE 품번 : TM01M-Z100-M021-0 수량 : 3 EA 발생기인(대분류) : 기구설계불량 발생기인(중분류) : 도면오류 발생기인(소분류) : 간섭불량 발생시점 : 공정 분류 : N/A 불량내용 : 볼트끼리 간섭 조치내용 : 자체처리 귀책 : 기구설계 PS : 2개 볼트 중 산 여유 있는 쪽에 한 치수 작은 볼트 사용. 이주환 조치 방안 회신해주시고, 입고 예정인 TM01M-Z100-M021-0_M 횡 전개하여 업체 수정 도면 전달 부탁드립니다. 1 16:51  1 16:52		
상세내용 Detail	#4,#5,#6 P/N: TM01M-Z100-M001-0, TM01M-Z100-M021-0 #7,#8,#9 P/N: TM01M-Z100-M001-0_M, TM01M-Z100-M021-0_M	Relocated the back plate mounting bolt holes. #4,#5,#6 P/N: TM01M-Z100-M001-1, TM01M-Z100-M021-1 #7,#8,#9 P/N: TM01M-Z100-M001-1_M, TM01M-Z100-M021-1_M	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

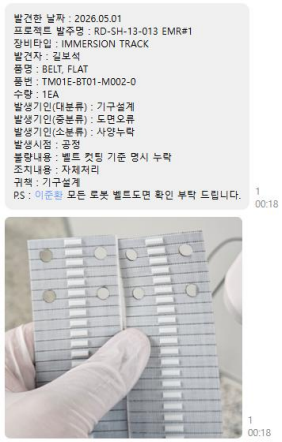
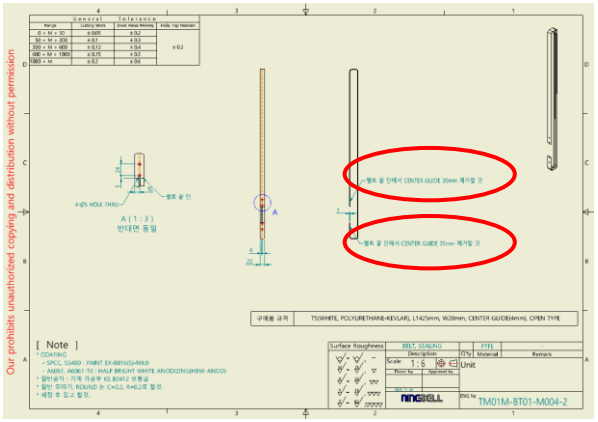
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

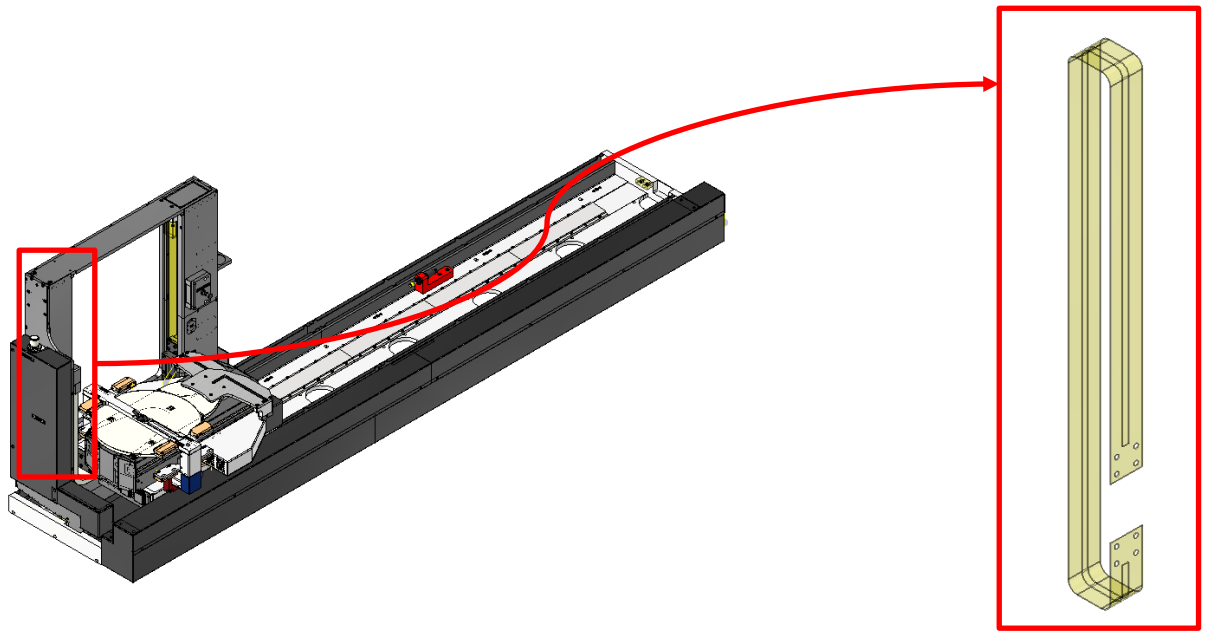
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #1, #12	적용 범위 Scope	■ 신규 발주분 (New order) □ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO. ECN NO.	N/A ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Missing specification for belt cutting criteria.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
			
상세내용 Detail	#1 P/N: TM01E-BT01-M002-0 #12 P/N: TM01I-BT01-M009-2	Added a note: "Remove 35mm of the CENTER GUIDE from the end of the belt." #1 P/N: TM01E-BT01-M002-1 #12 P/N: TM01I-BT01-M009-3	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

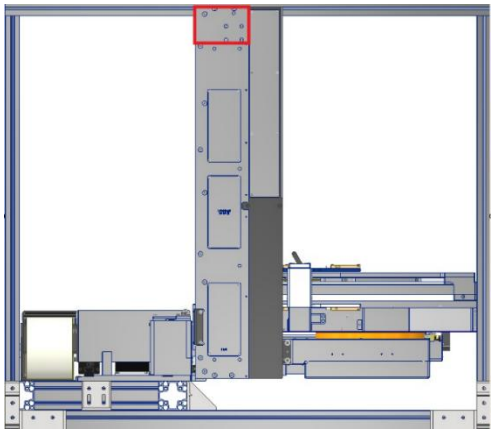
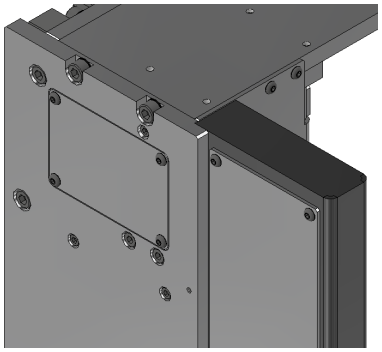
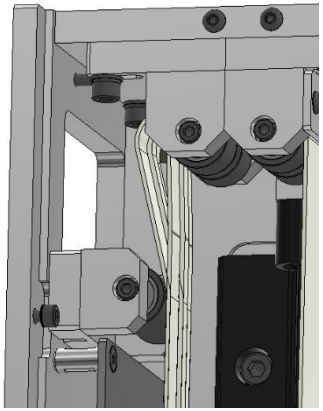
3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

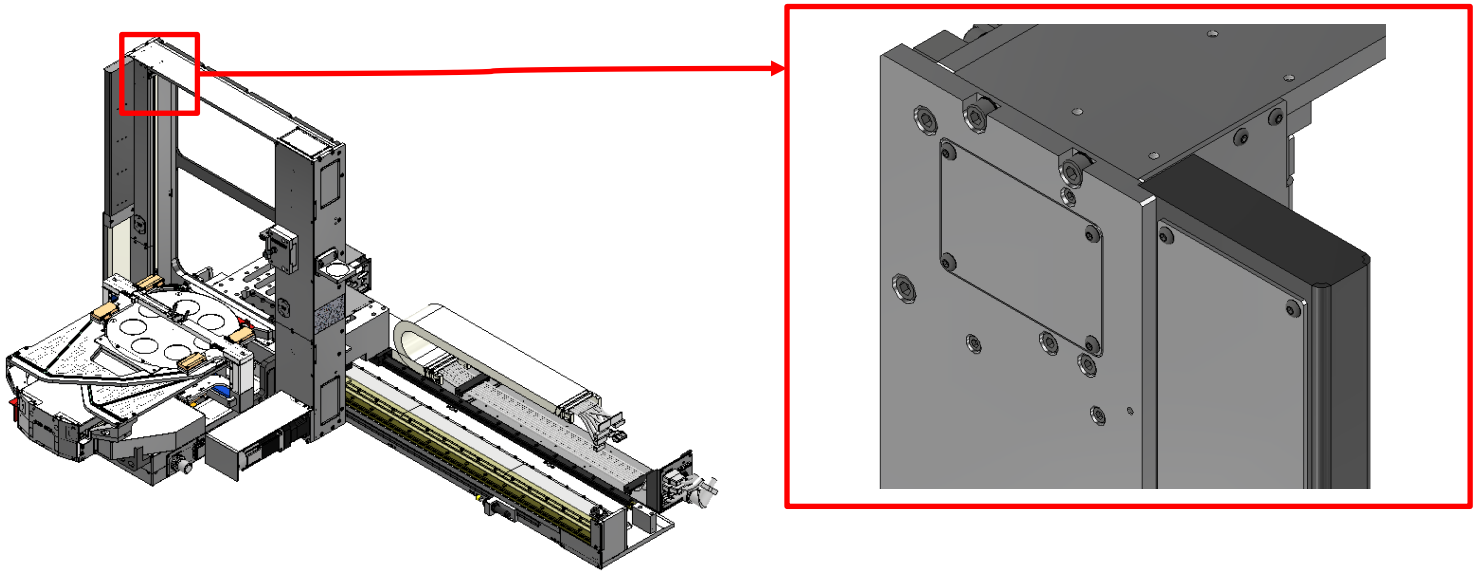
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2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-03-7813 IMMERSION TRACK PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #4 ~ #9	적용 범위 Scope	■ 신규 발주분 (New order) ■ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	Difficult to adjust the inner fastening bolts when adjusting the Z-axis tension.		
변경 사항 Changes	변경 전(Before)	변경 후(After)	
		 	
상세내용 Detail	#4,#5,#6 P/N: TM01M-Z100-M009-0, TM01M-Z400-M003-0 #7,#8,#9 P/N: TM01M-Z100-M009-0_M, TM01M-Z400-M003-0	Added a pocket to the Z-axis side plate (for tool access). #4,#5,#6 P/N: TM01M-Z100-L007-0, TM01M-Z100-M009-1, TM01M-Z400-M003-1 #7,#8,#9 P/N: TM01M-Z100-L007-0, TM01M-Z100-M009-1_M, TM01M-Z400-M003-1	

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / N	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	Y / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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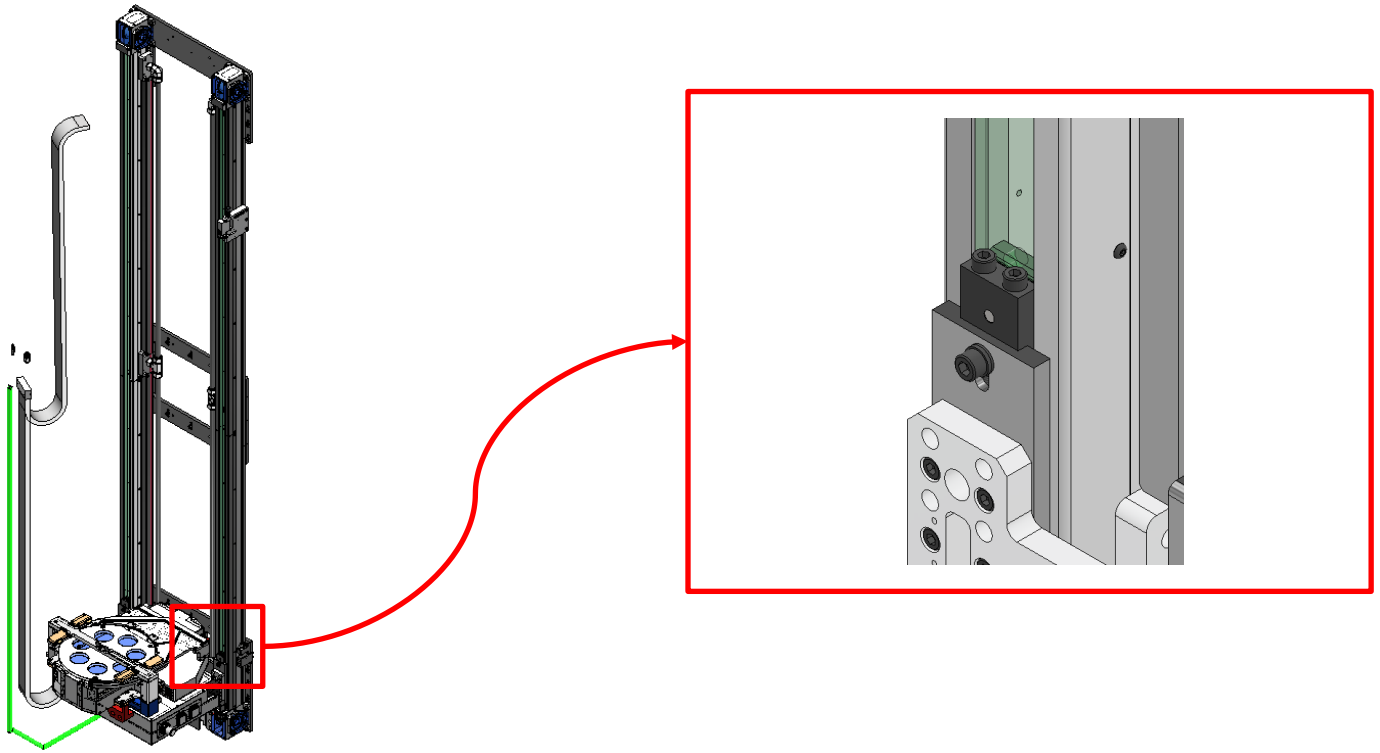
2. Engineering Change Notice

적용 장비 타입 Applicable project type	PO-2026-03-7813 IMMERSION TRACK PO-2026-02-3220 MISC-IMMERSION TRACK #1~#4 Application to robot #2, #3, #10, #11	적용 범위 Scope	■ 신규 발주분 (New order) ■ 기발주분 (Previous order) □ 기출하장비 (Existing shipped equipment)
변경 사유 Reason for change	□ 고객 요청 (Customer request) ■ 품질개선 (Quality improve) □ 원가절감 (Cost reduction) ■ 생산성 향상 (Increase productivity) □ 부품 단종 (Part discontinuation) ■ 기타 (ETC)	ECR NO.	N/A
		ECN NO.	ECN-D-2607-02
변경 상세 사유 Detailed reason for change	A thin gap occurs between the Z-axis timing belt and the side covers.		

변경 사항 Changes	변경 전(Before)	변경 후(After)
	2) ESR Z AXIS BELT ALIGN • 1. ALIGN 정렬이 직선이 될 수 없어 브라켓과 틈 발생	GE
상세내용 Detail	#2,#3 P/N: TM01S-Z110-M004-0, TM01S-Z110-M012-0, TM01S-Z110-P003-0, TM01S-Z110-P003-0_M, TM01S-Z110-P004-0, TM01S-Z110-P004-0_M, TM01S-Z110-P005-0, TM01S-Z110-P006-0 #10,#11 P/N: TM01S-Z110-M004-0_M, TM01S-Z110-M012-0, TM01S-Z110-P003-0, TM01S-Z110-P003-0_M, TM01S-Z110-P004-0, TM01S-Z110-P004-0_M, TM01S-Z110-P005-0, TM01S-Z110-P006-0	Redesigned to a cover-type structure that shields the timing belt. #2,#3 P/N: TM01S-Z110-M004-1, TM01S-Z110-M012-1, TM01S-Z110-P003-1, TM01S-Z110-P003-1_M, TM01S-Z110-P004-1, TM01S-Z110-P004-1_M, TM01S-Z110-P005-1, TM01S-Z110-P006-1 #10,#11 P/N: TM01S-Z110-M004-1_M, TM01S-Z110-M012-1, TM01S-Z110-P003-1, TM01S-Z110-P003-1_M, TM01S-Z110-P004-1, TM01S-Z110-P004-1_M, TM01S-Z110-P005-1, TM01S-Z110-P006-1

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2. Engineering Change Notice

전체 이미지 Full Image	 ※변경 위치 혹은 관련 위치 표시 ※Show changes or related positions

유효성 검토 결과 Test Result	검토 여부 Test or not	Y / (N)	유효성 검토 부서 Department	N/A	고객 승인 여부 Customer approva	(Y) / N
승인 날짜 Approval date	1. Ninebell : 2026 / 07 / 08 2. Customer : 2026 / 07 / 08		나인벨 승인 Ninebell approval	QC 2026.07.08 PASS	고객사 승인 Customer approval	Yes
적용 일자 Effective date	NEW PO		최초 적용 장비 Initially Applied Equipment	NEW PO		

3. Attachments

N/A

Check Sheet			
Request item		Date	Check Box
1	ACM Request	-	-
2	NB Response	26/06/30	O
3	ACM Confirm	26/07/08	O
4	Modification 3D, 2D		
5	ACM Confirm		

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